MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SHIFT REGISTER, DIGITAL, KEY CODE SJS

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic shift register module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

* 2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.164 failures per million operating hours for class I or 0.164 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be SJS.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/273-1 for class I and M28787/273-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

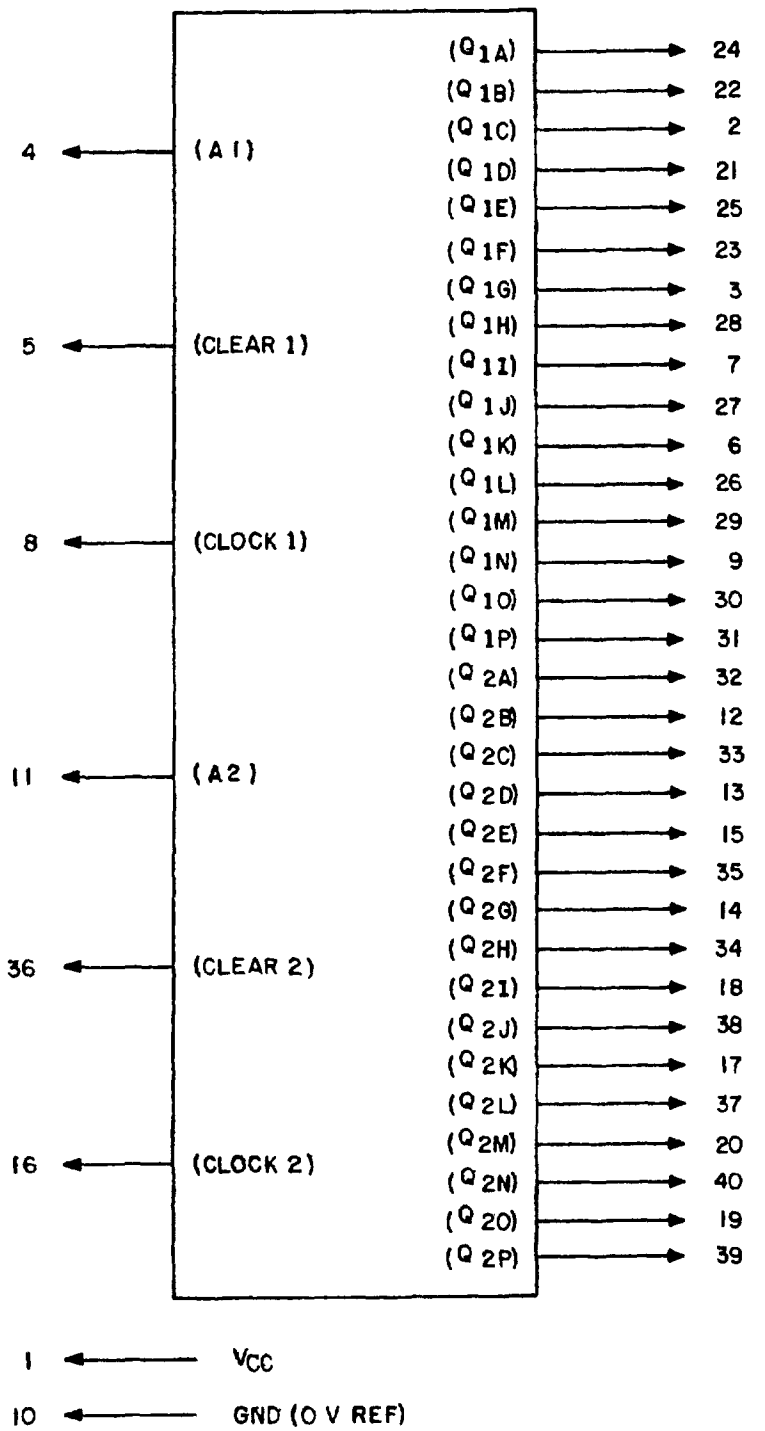
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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* FIGURE 1. SJS module function diagram, shift register.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B32	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
C1-C6	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
D1-D32	Output short circuit current	VI	X	-10.0	-27.5	-10.0	-27.5	-10.0	-27.5	-10.0	-27.5	-10.0	-27.5	mA
E1-E70	Functional	VII	X	← See table VII →										
F1-F6	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G32	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	800	mV
H1-H32	Output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1,J3,J4,J6	Input currnet, high level	XI	X	0	80	0	80	0	80	0	80	0	80	µA
J2, J5	Input current, high level	XI	X	-786	-1008	-786	-1008	-786	-1008	-786	-1008	-786	-1008	µA
K1, K4	Input current, low level	XII	X	-0.01	-1.73	-0.01	-1.73	-0.01	-1.73	-0.01	-1.73	-0.01	-1.73	mA
K2, K5	Input current, low level	XII	X	-1.56	-5.21	-1.56	-5.21	-1.56	-5.21	-1.56	-5.21	-1.56	-5.21	mA
K3, K6	Input current, low level	XII	X	-0.01	-3.45	-0.01	-3.45	-0.01	-3.45	-0.01	-3.45	-0.01	-3.45	mA
L1	Power supply current, low level	XIII	X	50	216	50	216	50	216	50	216	50	216	mA
L2	Power supply current, high level	XIII	X	50	216	50	216	50	216	50	216	50	216	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1-M32	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	5	85	5	85	5	90	5	85	5	90	ns
M33-M64	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	15	130	15	130	15	135	15	130	15	135	ns
N1-N32	Transition time (t_{THL}), C_L = 50 pF	XV	S	4	20	4	20	4	22	4	20	4	24	ns
N33-N64	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	5	30	5	30	5	35	5	35	5	35	ns
P1-P32	Propagation delay time (t_{PLH}), (clock to output), C_L = 50 pF	XVI	S	10	30	10	30	10	35	10	35	10	35	ns
P33-P64	Propagation delay time (t_{PLH}), (clock to output), C_L = 150 pF	XVI	Q	10	45	10	45	10	50	10	45	10	50	ns
Q1-Q32	Propagation delay time (t_{PHL}), (clock to output), C_L = 50 pF	XVII	S	10	37	10	37	10	42	10	37	10	42	ns
Q33-Q64	Propagation delay time (t_{PHL}), (clock to output), C_L = 150 pF	XVII	Q	10	47	10	47	10	52	10	47	10	52	ns
R1-R32	Propagation delay time (t_{PHL}), (clear to output), C_L = 50 pF	XVIII	S	10	42	10	42	10	47	10	42	10	47	ns
R33-R64	Propagation delay time (t_{PHL}), (clear to output), C_L = 150 pF	XVIII	Q	10	52	10	52	10	57	10	52	10	57	ns
S1-S32	Maxium shift frequency	XIX	S	72	88	72	88	72	88	72	88	72	88	ns

- - -

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T1-T32	Output crosstalk, low level	XX	Q	0	400	0	400	0	400	0	400	0	400	mV
U1-U32	Output crosstalk, high level	XXI	Q	0	400	0	400	0	400	0	400	0	400	mV
V	Power supply transient amplitude	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
W	Power supply filter capacitance	XXIII	Q	3.30	6.12	3.30	6.12	3.30	6.12	3.30	6.12	3.30	6.12	µF
X1-X6	Input capacitance	XXIV	Q	2	30	2	30	2	30	2	30	2	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition W, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through X shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded):	1,188	1,188	mW
Input voltage:			
Low level	0 min, 0.8 max	0 min, 0.8 max	V
High level	2.0 min, V_{CC} max	2.0 min, V_{CC} max	V

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	Class I	Class II	Unit
Input current (clear inputs):			
Low level (V_{IL} = 0.4 V)	-5.21 max	-5.21 max	mA
High level (V_{IH} = 2.4 V)	-1008 max	-1008 max	µA
Output voltage			
Low level (I_{OL} = 8.0 mA)	0.4 max	0.4 max	V
High level (I_{OH} = -400 µA	2.4 min	2.4 min	V
Output drive current (maximum capability)			
Low level (V_{OL} = 0.4 V)	8 max	8 max	mA
High level (V_{OH} = 2.4 V)	-400 max	-400 max	µA
Propagation delay time (clock to output)			
t_{PLH}			
C_L = 50 pF	10 min, 35 max	10 min, 35 max	ns
C_L = 150 pF	10 min, 50 max	10 min, 50 max	ns
t_{PHL}			
C_L = 50 pF	10 min, 42 max	10 min, 42 max	ns
C_L = 150 pF	10 min, 52 max	10 min, 52 max	ns
Propagation delay time (clear to output)			
t_{PHL}			
C_L = 50 pF	10 min, 47 max	10 min, 47 max	ns
C_L = 150 pF	10 min, 57 max	10 min, 57 max	ns
Clock input			
Frequency	25 max	25 max	MHz
Pulse width	20 min	20 min	ns
Clear pulse width	20 min	20 min	ns
Hold time			
t_{HLH}, t_{HHL}	5 min	5 min	ns
Setup time			
t_{SLH}, t_{SHL}	15 min	15 min	ns

* 6.5 Module function. The module contains two independent 16-bit serial to parallel shift registers with asynchronous clear capability and with all parallel outputs available. Each shift register has one serial input.

6.5.1 Input to output relationship. Each shift register has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/-6/

Inputs			Outputs		
Clear	Clock	Input	Q_A	Q_B	Q_P
L	X	X	L	L	L
H	L	X	Q_{AO}	Q_{BO}	Q_{PO}
H		H	H	Q_{AN}	Q_{ON}
H		L	L	Q_{AN}	Q_{ON}

1/ H = High level (steady state).
2/ L = Low level (steady state).
3/ X = Irrelevant (any input, including transitions).
4/ = Transition from low-to-high input.
5/ Q_{AO}, Q_{BO}, Q_{PO} = The level of Q_A, Q_B, or Q_P respectively, before the indicated steady-state input conditions were established.
6/ Q_{AN}, Q_{ON} = The level of Q_A or Q_O before the most recent transition of the clock, indicates a one bit shift.

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6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

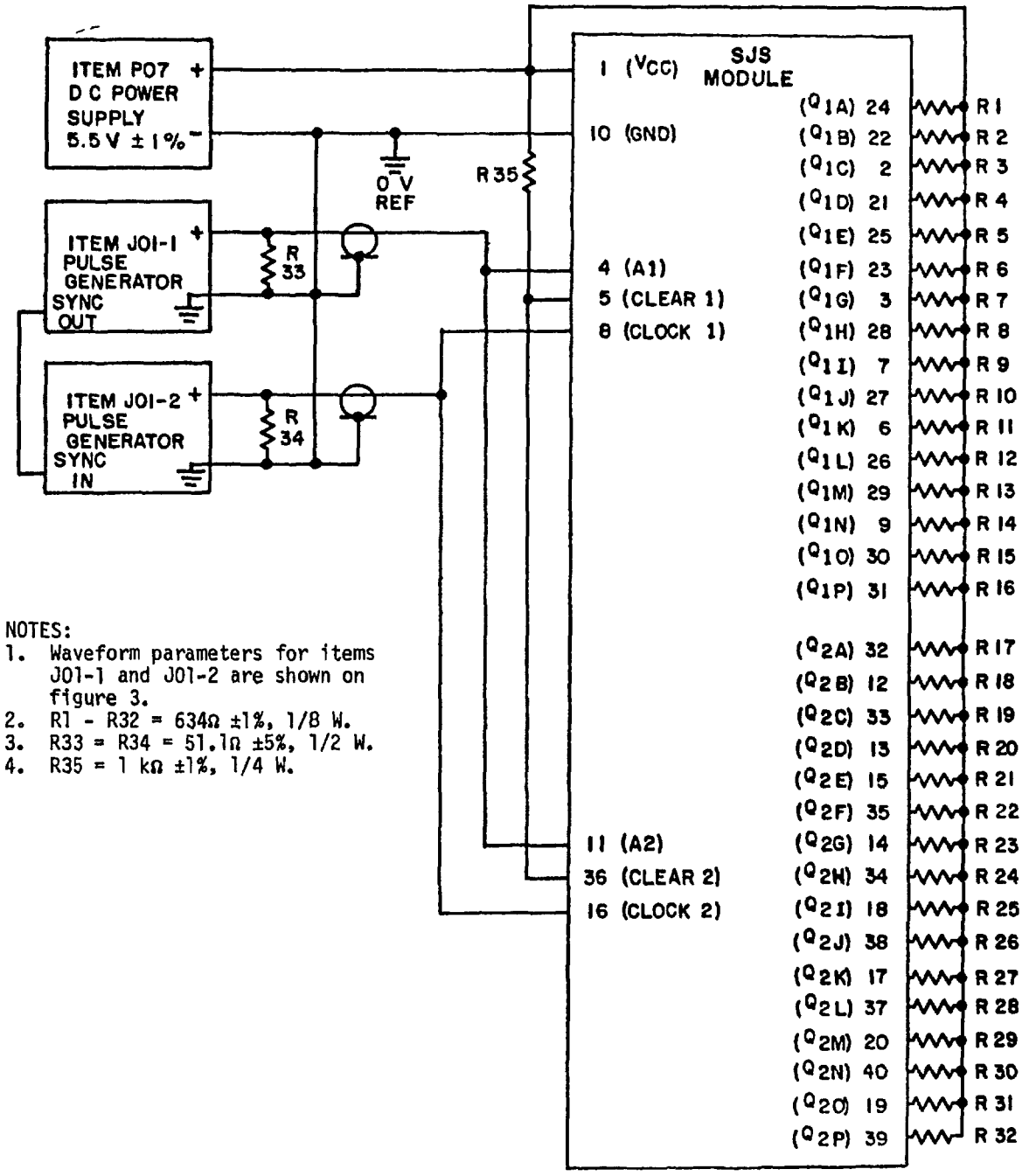
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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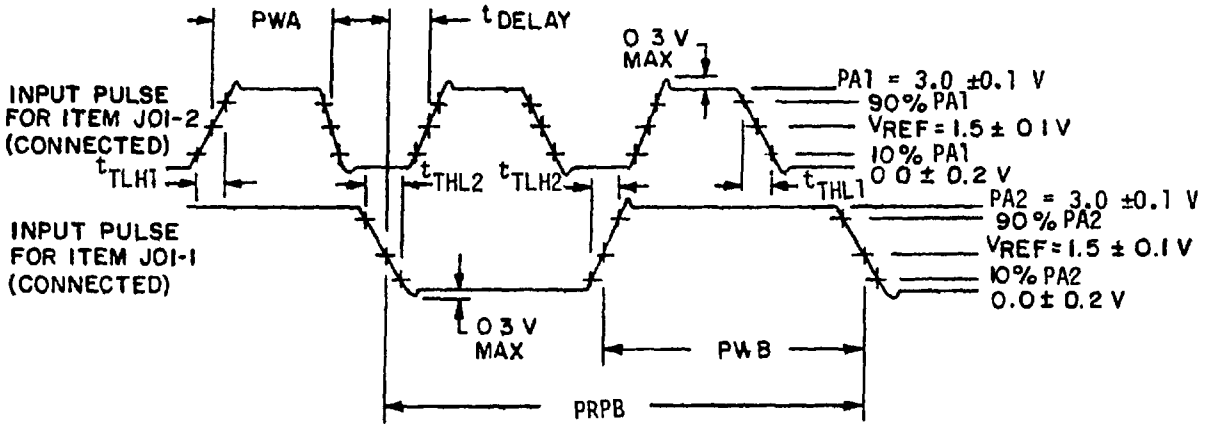
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NOTES:
1. Waveform parameters for items J01-1 and J01-2 are shown on figure 3.
2. R1 - R32 = 634Ω ±1%, 1/8 W.
3. R33 = R34 = 51.1Ω ±5%, 1/2 W.
4. R35 = 1 kΩ ±1%, 1/4 W.

FIGURE 2. Operating temperature and life test circuit.

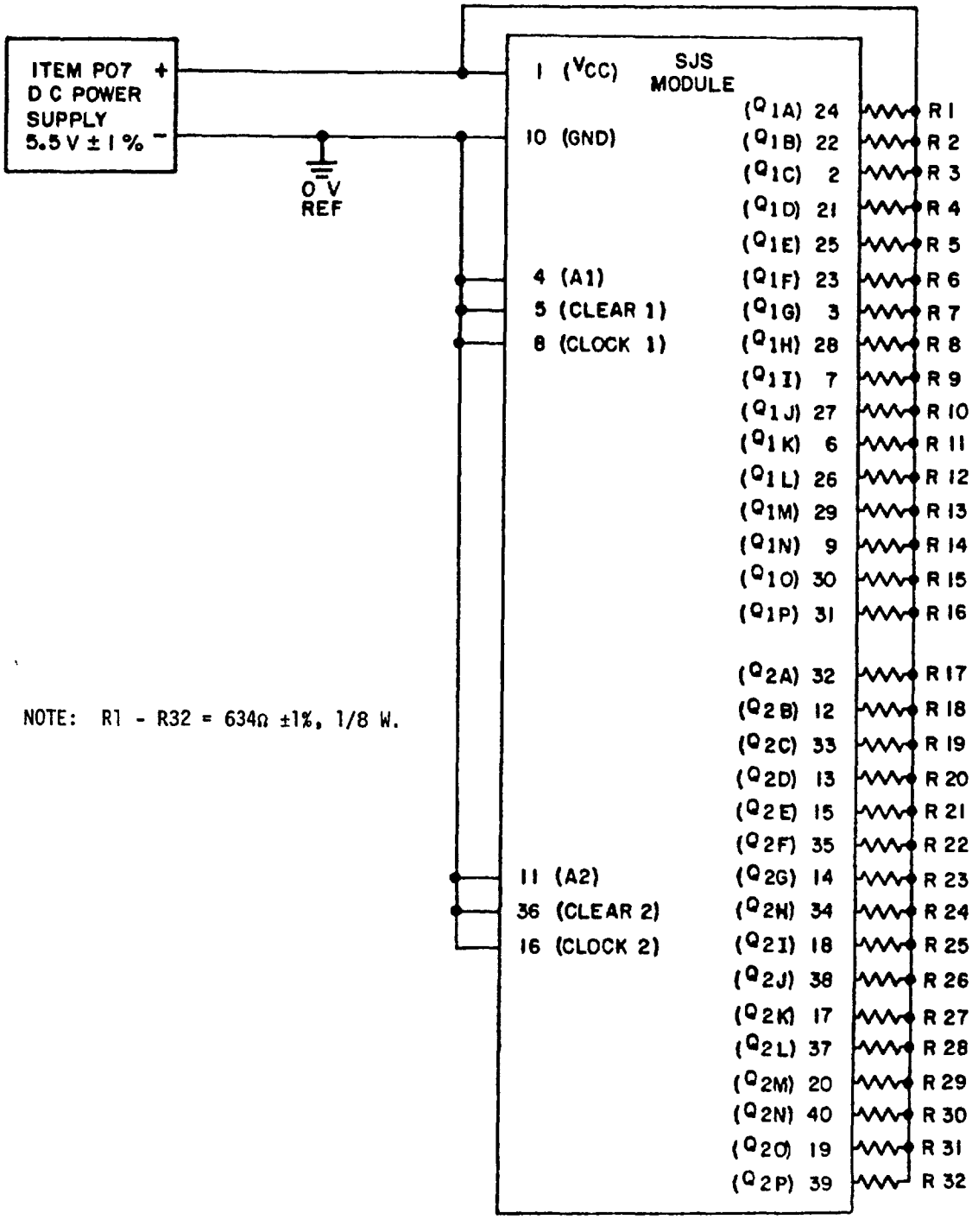
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NOTES:
1. Input pulse parameters for item J01-2:
   PWA = 250 ±25 ns.
   $t_{TLH1} = t_{THL1} = 50 \pm 10$ ns.
   Put item J01-2 in the double pulse mode.
   $t_{DELAY} = 200 \pm 20$ ns.
2. Input pulse parameters for item J01-1:
   PWB = 500 ±50 ns.
   $t_{TLH2} = t_{THL2} = 50 \pm 10$ ns.
   PRPB = 1,000 ±100 ns.
3. All voltages are referenced to ground.

* FIGURE 3. Operating temperature and life test circuit waveforms for figure 2.

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NOTE: R1 - R32 = 634Ω ±1%, 1/8 W.

FIGURE 4. Component temperature test circuit.

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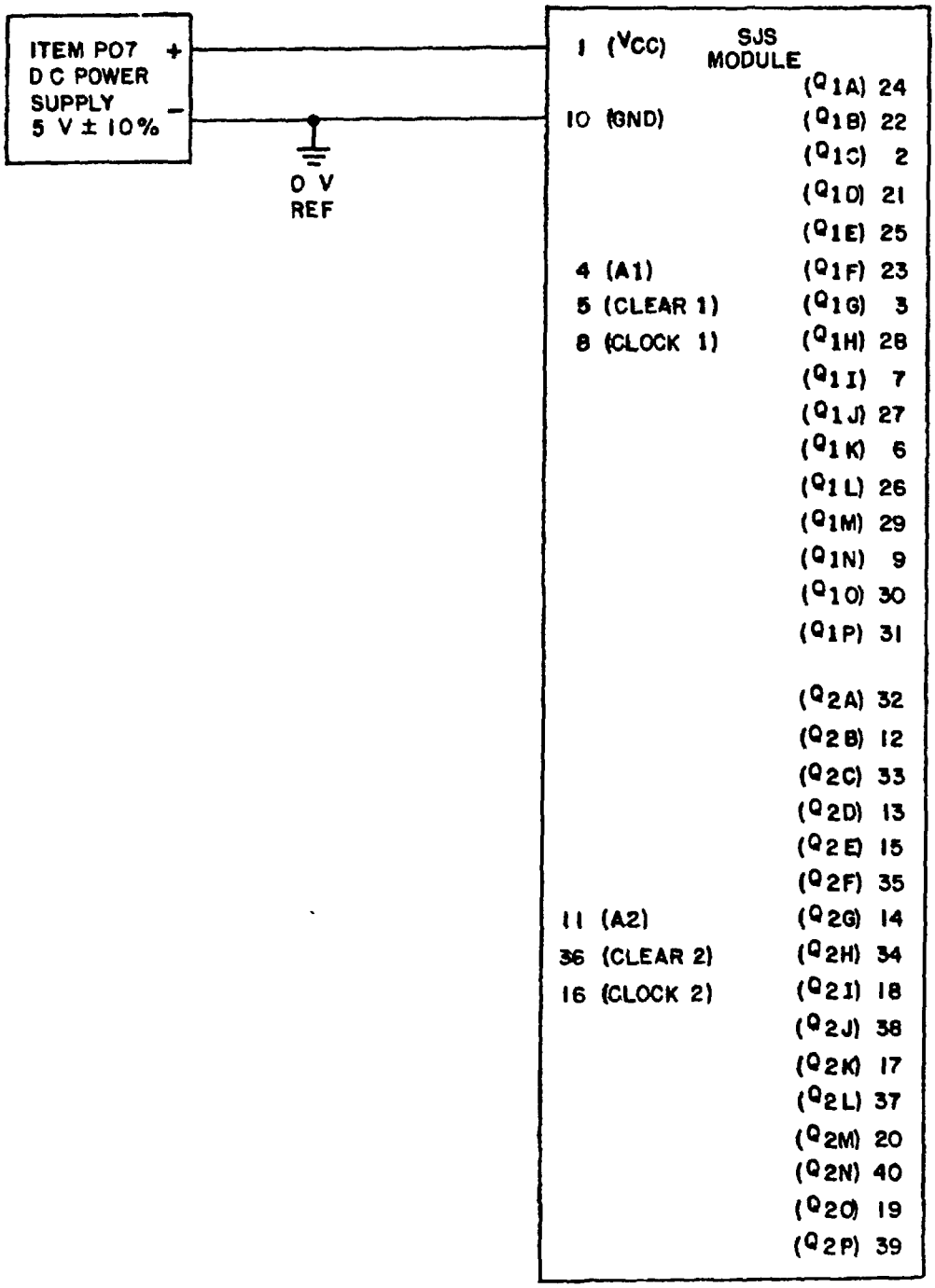


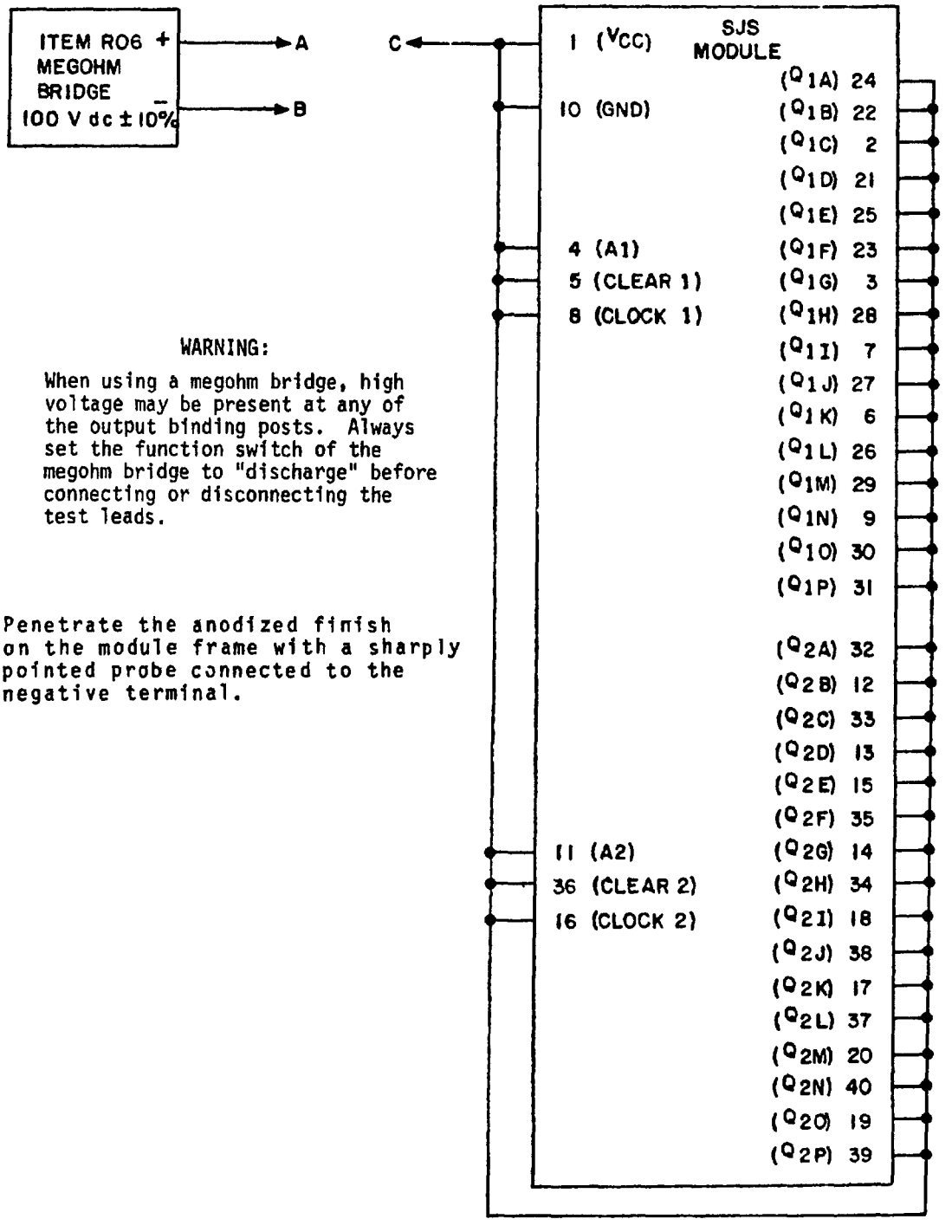
FIGURE 5. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Module frame	10	∞	MΩ	6

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Isolation test circuit for table III.

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*TABLE IV. V_{OUT} (low-loaded) test.

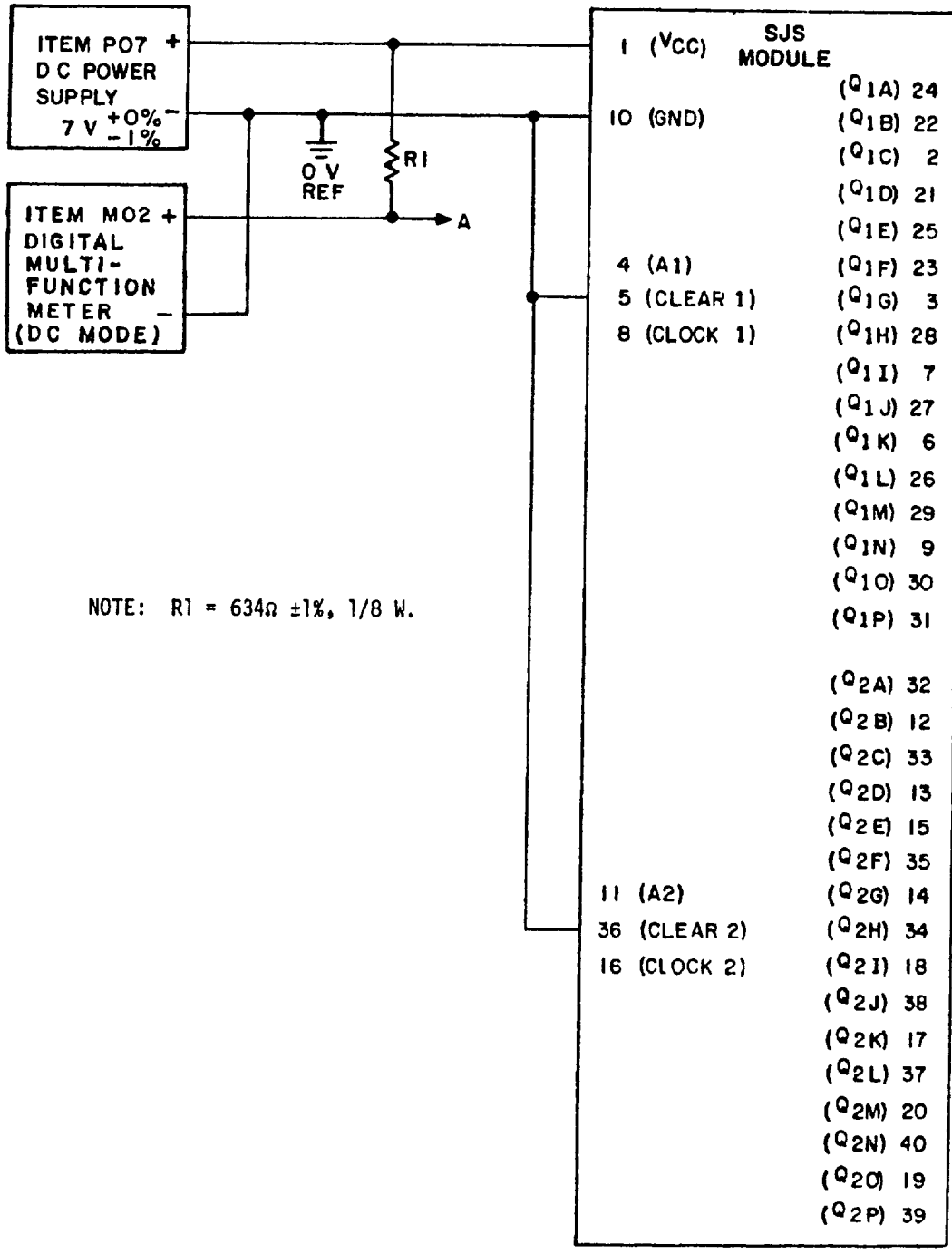
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	Q_{1A} V_{OUT} (low loaded)	24	10	800	mV	7
B2	Q_{1B} "	22	"	"	"	"
B3	Q_{1C} "	2	"	"	"	"
B4	Q_{1D} "	21	"	"	"	"
B5	Q_{1E} "	25	"	"	"	"
B6	Q_{1F} "	23	"	"	"	"
B7	Q_{1G} "	3	"	"	"	"
B8	Q_{1H} "	28	"	"	"	"
B9	Q_{1I} "	7	"	"	"	"
B10	Q_{1J} "	27	"	"	"	"
B11	Q_{1K} "	6	"	"	"	"
B12	Q_{1L} "	26	"	"	"	"
B13	Q_{1M} "	29	"	"	"	"
B14	Q_{1N} "	9	"	"	"	"
B15	Q_{1O} "	30	"	"	"	"
B16	Q_{1P} "	31	"	"	"	"
B17	Q_{2A} "	32	"	"	"	"
B18	Q_{2B} "	12	"	"	"	"
B19	Q_{2C} "	33	"	"	"	"

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*TABLE IV. V_{OUT} (low-loaded) test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B20	Q_{2D} V_{OUT} (low loaded)	13	10	800	mV	7
B21	Q_{2E} "	15	"	"	"	"
B22	Q_{2F} "	35	"	"	"	"
B23	Q_{2G} "	14	"	"	"	"
B24	Q_{2H} "	34	"	"	"	"
B25	Q_{2I} "	18	"	"	"	"
B26	Q_{2J} "	38	"	"	"	"
B27	Q_{2K} "	17	"	"	"	"
B28	Q_{2L} "	37	"	"	"	"
B29	Q_{2M} "	20	"	"	"	"
B30	Q_{2N} "	40	"	"	"	"
B31	Q_{2O} "	19	"	"	"	"
B32	Q_{2P} "	39	"	"	"	"

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* FIGURE 7. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	A1 Input breakdown current	5,8,11,36,16	4	0	1	mA	8
C2	CLEAR 1 "	4,8,11,36,16	5	"	"	"	"
C3	CLOCK 1 "	4,5,11,36,16	8	"	"	"	"
C4	A2 "	4,5,8,36,16	11	"	"	"	"
C5	CLEAR 2 "	4,5,8,11,16	36	"	"	"	"
C6	CLOCK 2 "	4,5,8,11,36	16	"	"	"	"

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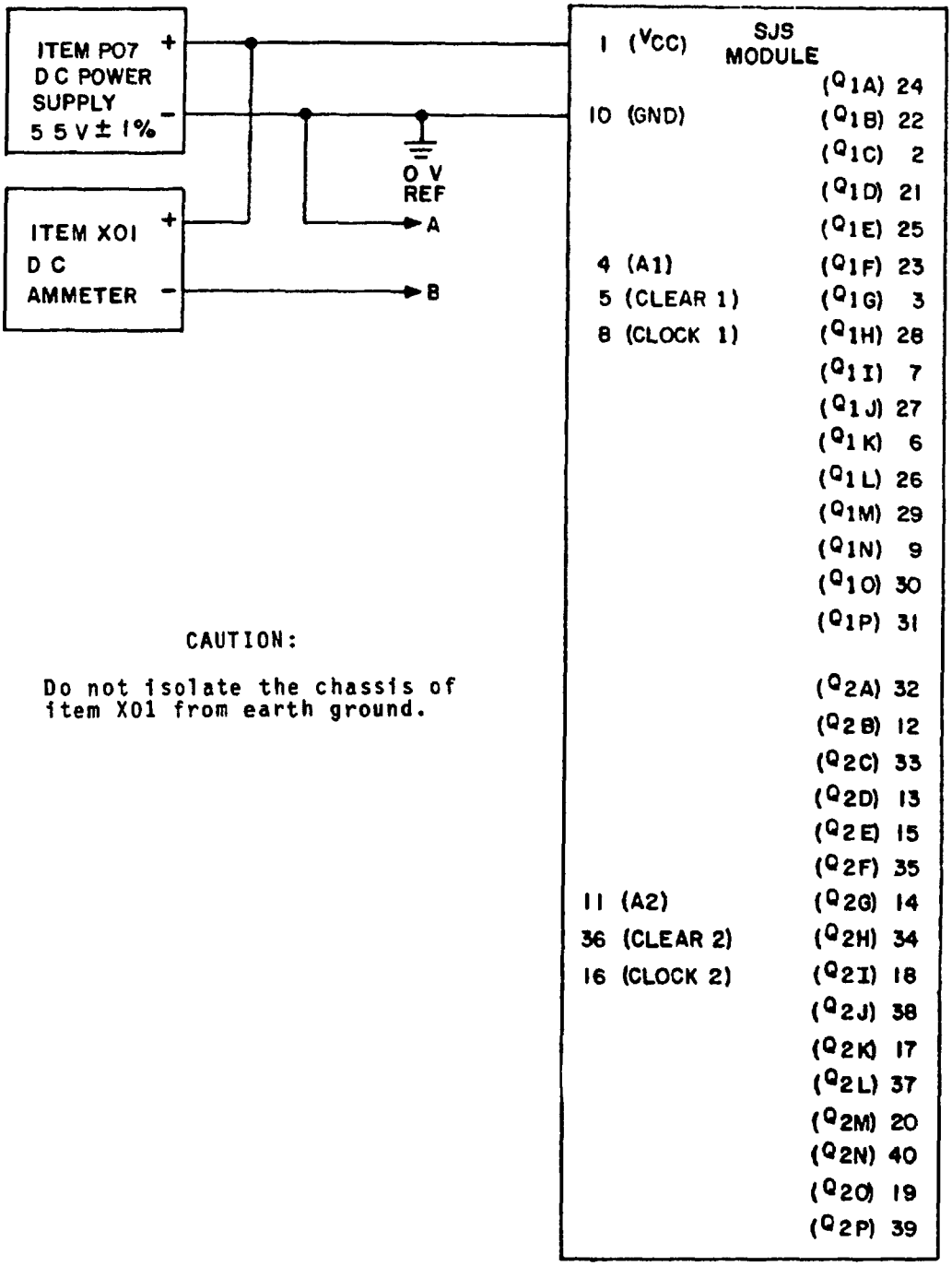


FIGURE 8. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	Q_{1A} Output short circuit current	24	-10.0	-27.5	mA	9,10
D2	Q_{1B} "	22	"	"	"	"
D3	Q_{1C} "	2	"	"	"	"
D4	Q_{1D} "	21	"	"	"	"
D5	Q_{1E} "	25	"	"	"	"
D6	Q_{1F} "	23	"	"	"	"
D7	Q_{1G} "	3	"	"	"	"
D8	Q_{1H} "	28	"	"	"	"
D9	Q_{1I} "	7	"	"	"	"
D10	Q_{1J} "	27	"	"	"	"
D11	Q_{1K} "	6	"	"	"	"
D12	Q_{1L} "	26	"	"	"	"
D13	Q_{1M} "	29	"	"	"	"
D14	Q_{1N} "	9	"	"	"	"
D15	Q_{1O} "	30	"	"	"	"
D16	Q_{1P} "	31	"	"	"	"
D17	Q_{2A} "	32	"	"	"	"
D18	Q_{2B} "	12	"	"	"	"
D19	Q_{2C} "	33	"	"	"	"

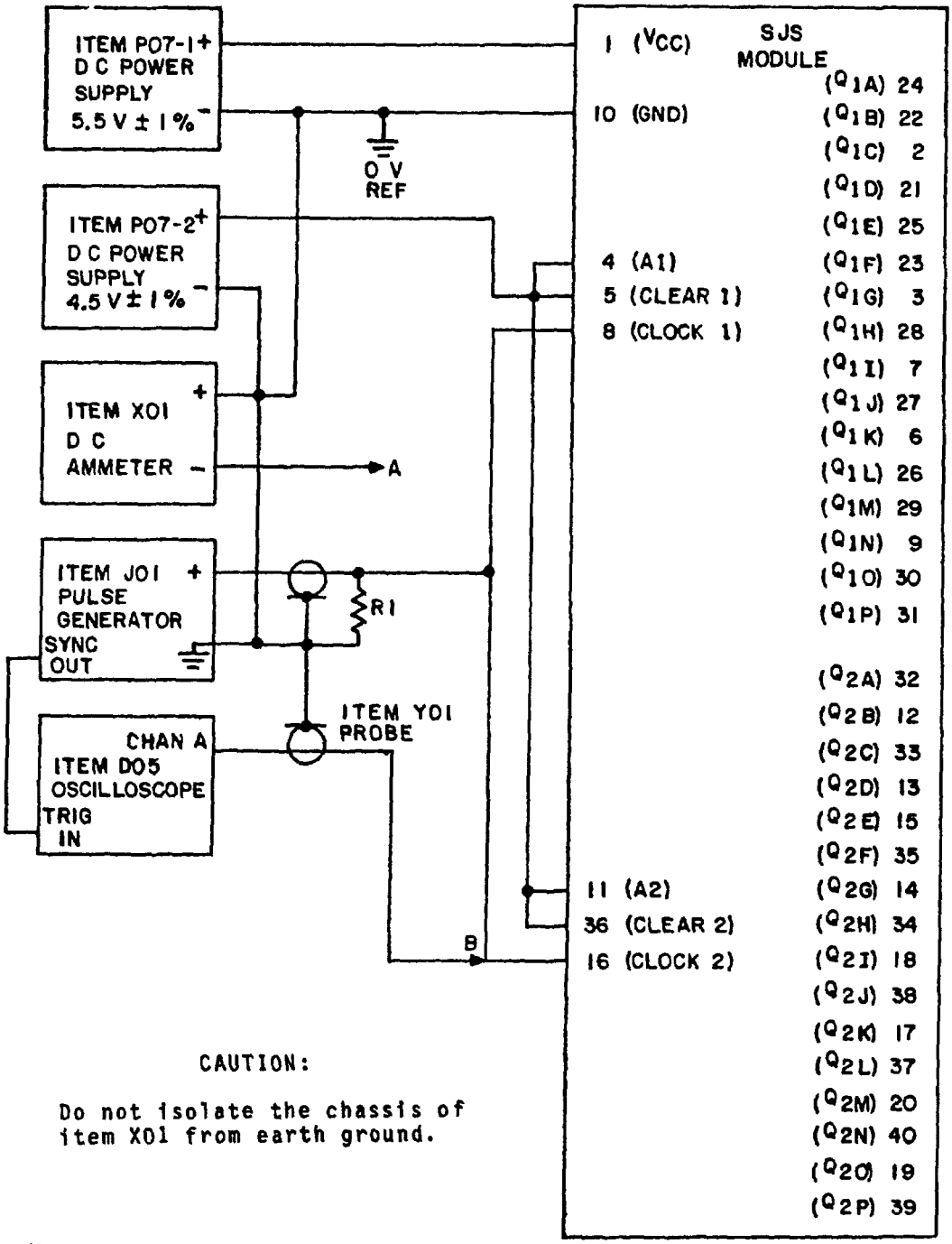
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TABLE VI. Output short circuit current test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D20	Q_{2D}	Output short circuit current	13	-10.0	-27.5	mA	9,10
D21	Q_{2E}	"	15	"	"	"	"
D22	Q_{2F}	"	35	"	"	"	"
D23	Q_{2G}	"	14	"	"	"	"
D24	Q_{2H}	"	34	"	"	"	"
D25	Q_{2I}	"	18	"	"	"	"
D26	Q_{2J}	"	38	"	"	"	"
D27	Q_{2K}	"	17	"	"	"	"
D28	Q_{2L}	"	37	"	"	"	"
D29	Q_{2M}	"	20	"	"	"	"
D30	Q_{2N}	"	40	"	"	"	"
D31	Q_{2O}	"	19	"	"	"	"
D32	Q_{2P}	"	39	"	"	"	"

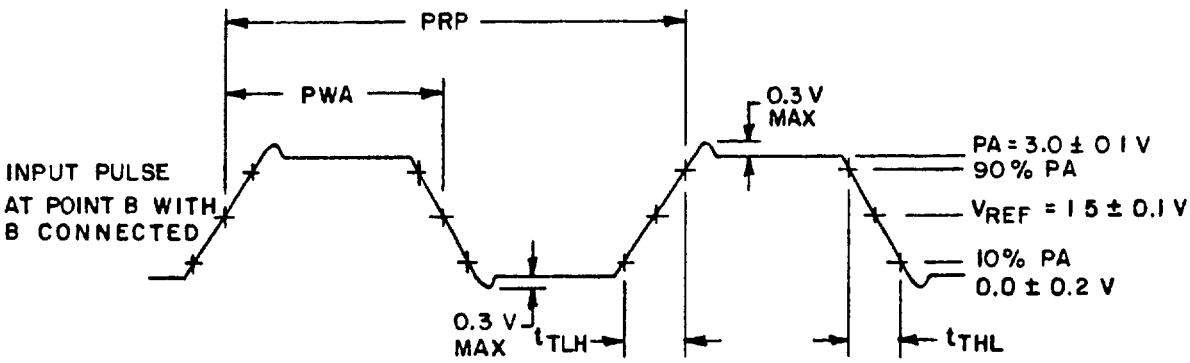
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. Before connecting point A, allow a minimum of 17 clock pulses. Before making any measurements, disable the clock.
3. When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 1 second.

FIGURE 9. Output short circuit current test circuit for table VI.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL}$ = 50 ±25 ns.
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns.

* FIGURE 10. Output short circuit current test circuit waveform for table VI.

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TABLE VII. Functional test. 1/,2/

Test condition	Parameter	Connect			No. of pulses	Outputs																Test figure
		A to pin	B to pin	C to pin		Q_{1A}	Q_{1B}	Q_{1C}	Q_{1D}	Q_{1E}	Q_{1F}	Q_{1G}	Q_{1H}	Q_{1I}	Q_{1J}	Q_{1K}	Q_{1L}	Q_{1M}	Q_{1N}	Q_{1O}	Q_{1P}	
E1	Functional	4	5	8	1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	11,12
E2	"	4,5	8	Open	0	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
E3	"	"	Open	8	1	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
E4	"	"	"	"	1	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
E5	"	"	"	"	1	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	"
E6	"	"	"	"	1	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	"
E7	"	"	"	"	1	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	"
E8	"	"	"	"	1	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	"
E9	"	"	"	"	1	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	"
E10	"	"	"	"	1	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	"
E11	"	"	"	"	1	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	"
E12	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"
E13	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	"
E14	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	"
E15	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	"
E16	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	"
E17	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	"
E18	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"

See footnotes at end of table.

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TABLE VII. Functional test - Continued. 1/,2/

Test condition	Parameter	Connect			No. of pulses	Outputs																Test figure
		A to pin	B to pin	C to pin		Q_{1A}	Q_{1B}	Q_{1C}	Q_{1D}	Q_{1E}	Q_{1F}	Q_{1G}	Q_{1H}	Q_{1I}	Q_{1J}	Q_{1K}	Q_{1L}	Q_{1M}	Q_{1N}	Q_{1O}	Q_{1P}	
E19	Functional	5	4	8	1	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	11,12
E20	"	"	"	"	1	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
E21	"	"	"	"	1	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	"
E22	"	"	"	"	1	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	"
E23	"	"	"	"	1	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	"
E24	"	"	"	"	1	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	"
E25	"	"	"	"	1	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	"
E26	"	"	"	"	1	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	"
E27	"	"	"	"	1	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	"
E28	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"
E29	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	"
E30	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	"
E31	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	"
E32	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	H	"
E33	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	"
E34	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
E35	"	Open	5,4	"	1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"

See footnotes at end of table.

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TABLE VII. Functional test - Continued. 1/,2/

Test condition	Parameter	Connect			No. of pulses	Outputs																Test figure
		A to pin	B to pin	C to pin		Q_{2A}	Q_{2B}	Q_{2C}	Q_{2D}	Q_{2E}	Q_{2F}	Q_{2G}	Q_{2H}	Q_{2I}	Q_{2J}	Q_{2K}	Q_{2L}	Q_{2M}	Q_{2N}	Q_{2O}	Q_{2P}	
E36	Functional	11	36	16	1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	11,12
E37	"	11.36	16	Open	0	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
E38	"	"	Open	16	1	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
E39	"	"	"	"	1	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
E40	"	"	"	"	1	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	"
E41	"	"	"	"	1	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	"
E42	"	"	"	"	1	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	"
E43	"	"	"	"	1	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	"
E44	"	"	"	"	1	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	L	"
E45	"	"	"	"	1	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	L	"
E46	"	"	"	"	1	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	L	"
E47	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"
E48	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	"
E49	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	"
E50	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	"
E51	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	"
E52	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	"
E53	"	"	"	"	1	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"

See footnotes at end of table.

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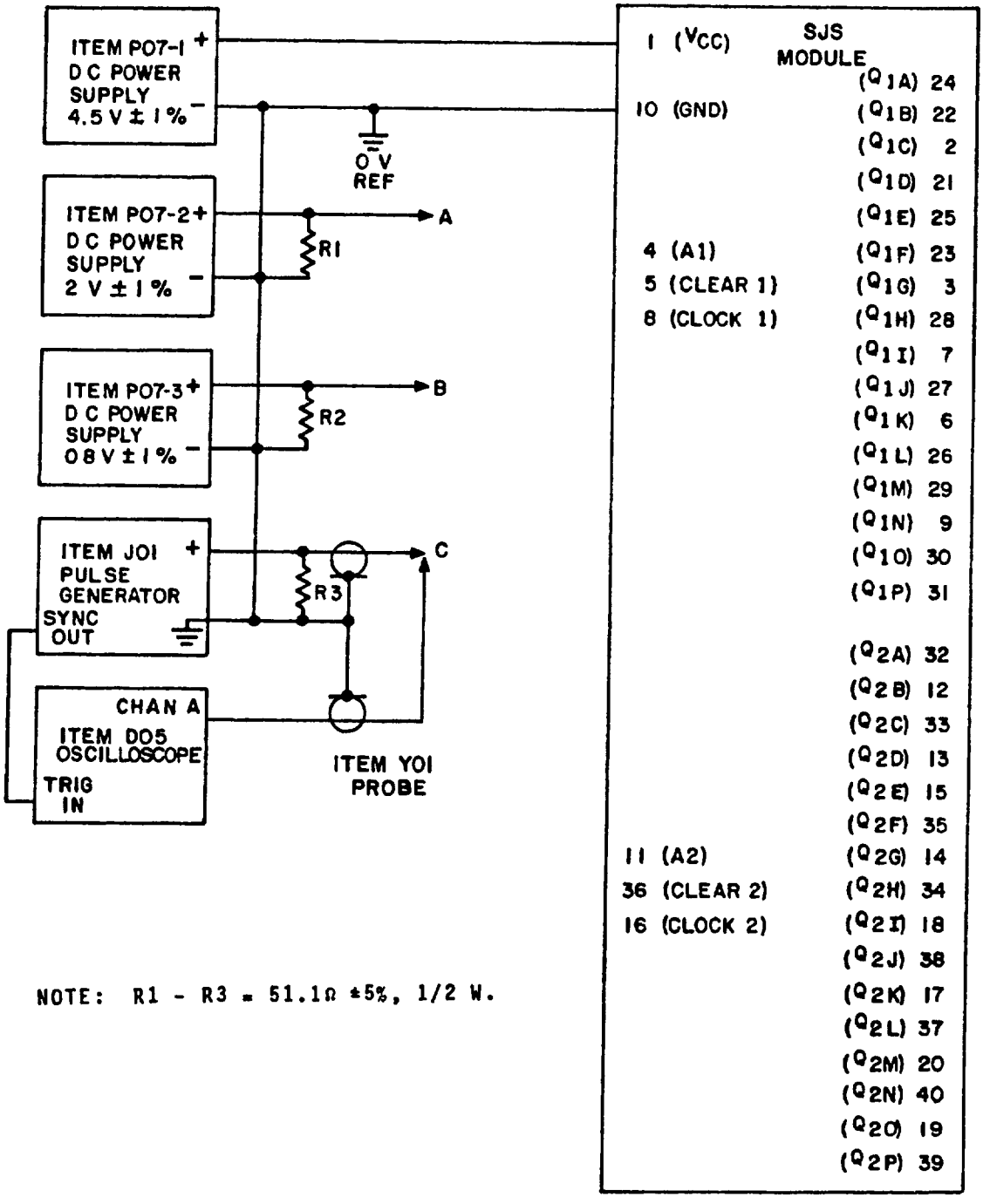
TABLE VII. Functional test - Continued. 1/.2/

Test condition	Parameter	Connect			No. of pulses	Outputs																Test figure
		A to pin	B to pin	C to pin		Q_{2A}	Q_{2B}	Q_{2C}	Q_{2D}	Q_{2E}	Q_{2F}	Q_{2G}	Q_{2H}	Q_{2I}	Q_{2J}	Q_{2K}	Q_{2L}	Q_{2M}	Q_{2N}	Q_{2O}	Q_{2P}	
E54	Functional	36	11	16	1	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	11,12
E55	"	"	"	"	1	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
E56	"	"	"	"	1	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	"
E57	"	"	"	"	1	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	"
E58	"	"	"	"	1	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	"
E59	"	"	"	"	1	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	"
E60	"	"	"	"	1	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	"
E61	"	"	"	"	1	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	"
E62	"	"	"	"	1	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	"
E63	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"
E64	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	"
E65	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	"
E66	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	L	L	H	H	H	"
E67	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	H	"
E68	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	"
E69	"	"	"	"	1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"
E70	"	Open	36,11	"	1	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	"

1/ L = Low level.

2/ H = High level.

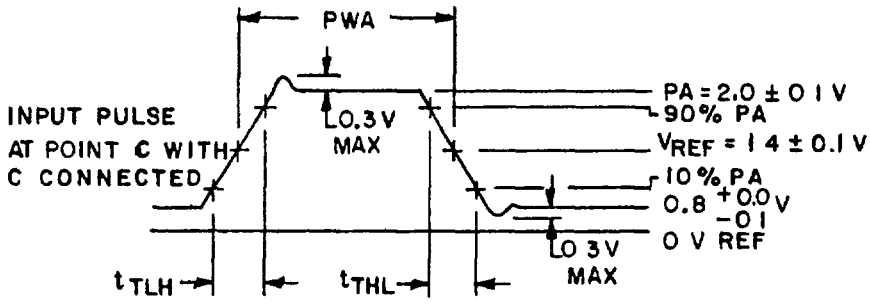
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NOTE: R1 - R3 = 51.1Ω ±5%, 1/2 W.

FIGURE 11. Functional test circuit for table VII.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 50 \pm 25$ ns.
PWA = 500 ±100 ns.
Put item J01 in the single pulse mode.

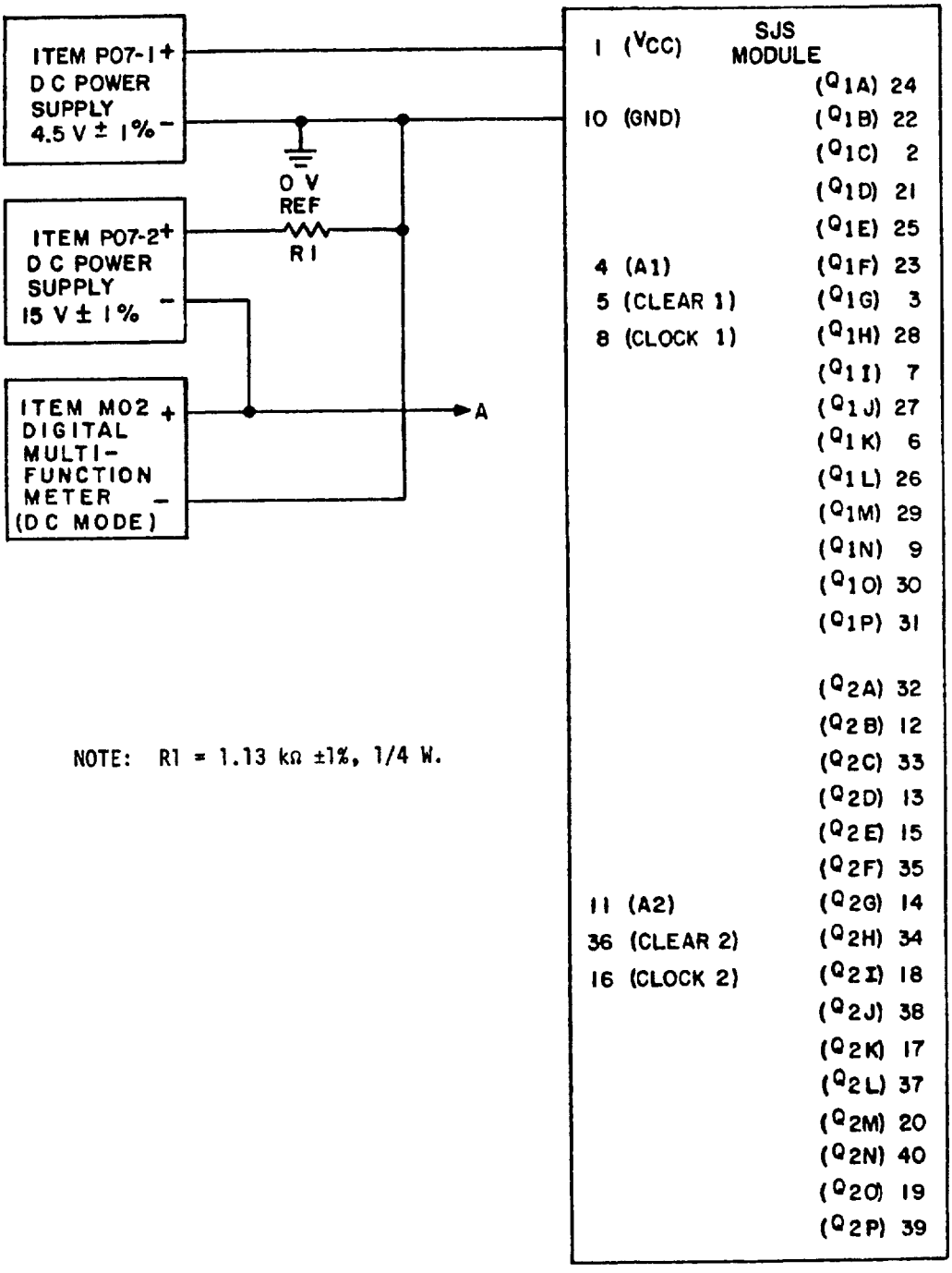
* FIGURE 12. Functional test circuit waveform for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	A1	Input clamp diode voltage	4	-0.4	-1.5	V	13
F2	CLEAR 1	"	5	"	"	"	"
F3	CLOCK 1	"	8	"	"	"	"
F4	A2	"	11	"	"	"	"
F5	CLEAR 2	"	36	"	"	"	"
F6	CLOCK 2	"	16	"	"	"	"

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NOTE: R1 = 1.13 kΩ ±1%, 1/4 W.

* FIGURE 13. Input clamp diode voltage test circuit for table VIII.

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*TABLE IX. Output voltage (low-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	Q_{1A}	Output voltage, low level	5, 36	Open	24	Open	0	400	mV	14, 15
G2	Q_{1B}	"	"	"	22	"	"	"	"	"
G3	Q_{1C}	"	"	"	2	"	"	"	"	"
G4	Q_{1D}	"	"	"	21	"	"	"	"	"
G5	Q_{1E}	"	"	"	25	"	"	"	"	"
G6	Q_{1F}	"	"	"	23	"	"	"	"	"
G7	Q_{1G}	"	"	"	3	"	"	"	"	"
G8	Q_{1H}	"	"	"	28	"	"	"	"	"
G9	Q_{1I}	"	"	"	7	"	"	"	"	"
G10	Q_{1J}	"	"	"	27	"	"	"	"	"
G11	Q_{1K}	"	"	"	6	"	"	"	"	"
G12	Q_{1L}	"	"	"	26	"	"	"	"	"
G13	Q_{1M}	"	"	"	29	"	"	"	"	"
G14	Q_{1N}	"	"	"	9	"	"	"	"	"
G15	Q_{1O}	"	"	"	30	"	"	"	"	"
G16	Q_{1P}	"	"	"	31	"	"	"	"	"
G17	Q_{2A}	"	"	"	32	"	"	"	"	"
G18	Q_{2B}	"	"	"	12	"	"	"	"	"

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*TABLE IX. Output voltage (low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
G19	Q_{2C}	Output voltage, low level	5, 36	Open	33	Open	0	400	mV	14, 15
G20	Q_{2D}	"	"	"	13	"	"	"	"	"
G21	Q_{2E}	"	"	"	15	"	"	"	"	"
G22	Q_{2F}	"	"	"	35	"	"	"	"	"
G23	Q_{2G}	"	"	"	14	"	"	"	"	"
G24	Q_{2H}	"	"	"	34	"	"	"	"	"
G25	Q_{2I}	"	"	"	18	"	"	"	"	"
G26	Q_{2J}	"	"	"	38	"	"	"	"	"
G27	Q_{2K}	"	"	"	17	"	"	"	"	"
G28	Q_{2L}	"	"	"	37	"	"	"	"	"
G29	Q_{2M}	"	"	"	20	"	"	"	"	"
G30	Q_{2N}	"	"	"	40	"	"	"	"	"
G31	Q_{2O}	"	"	"	19	"	"	"	"	"
G32	Q_{2P}	"	"	"	39	"	"	"	"	"

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*TABLE X. Output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	Q_{1A} Output voltage, high level	Open	4,5,11,36	24	8,16	2.4	4.5	V	14, 15
H2	Q_{1B} "	"	"	22	"	"	"	"	"
H3	Q_{1C} "	"	"	2	"	"	"	"	"
H4	Q_{1D} "	"	"	21	"	"	"	"	"
H5	Q_{1E} "	"	"	25	"	"	"	"	"
H6	Q_{1F} "	"	"	23	"	"	"	"	"
H7	Q_{1G} "	"	"	3	"	"	"	"	"
H8	Q_{1H} "	"	"	28	"	"	"	"	"
H9	Q_{1I} "	"	"	7	"	"	"	"	"
H10	Q_{1J} "	"	"	27	"	"	"	"	"
H11	Q_{1K} "	"	"	6	"	"	"	"	"
H12	Q_{1L} "	"	"	26	"	"	"	"	"
H13	Q_{1M} "	"	"	29	"	"	"	"	"
H14	Q_{1N} "	"	"	9	"	"	"	"	"
H15	Q_{1O} "	"	"	30	"	"	"	"	"
H16	Q_{1P} "	"	"	31	"	"	"	"	"
H17	Q_{2A} "	"	"	32	"	"	"	"	"
H18	Q_{2B} "	"	"	12	"	"	"	"	"

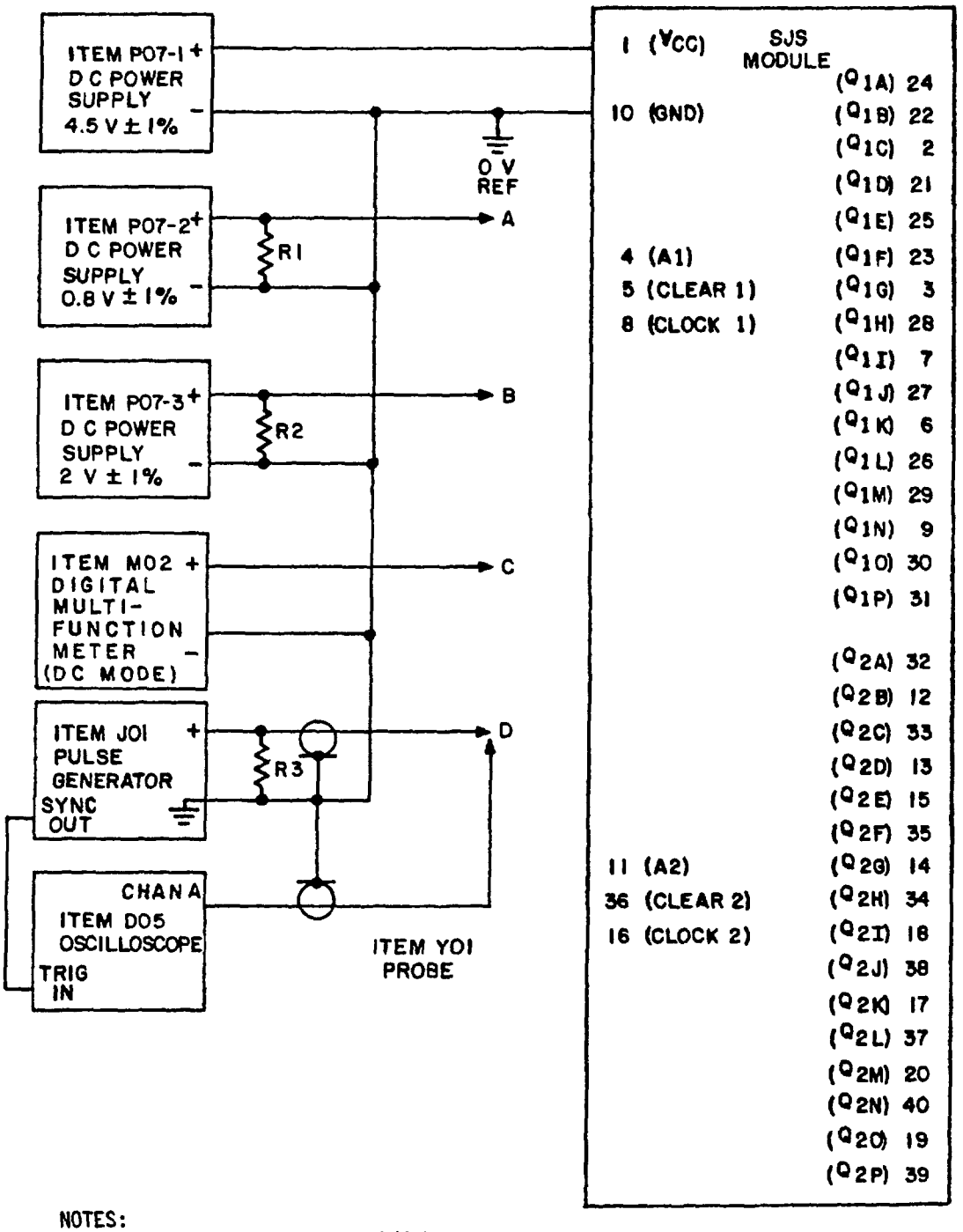
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*TABLE X. Output voltage (high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
H19	Q_{2C}	Output voltage, high level	Open	4,5,11,36	33	8,16	2.4	4.5	V	14, 15
H20	Q_{2D}	"	"	"	13	"	"	"	"	"
H21	Q_{2E}	"	"	"	15	"	"	"	"	"
H22	Q_{2F}	"	"	"	35	"	"	"	"	"
H23	Q_{2G}	"	"	"	14	"	"	"	"	"
H24	Q_{2H}	"	"	"	34	"	"	"	"	"
H25	Q_{2I}	"	"	"	18	"	"	"	"	"
H26	Q_{2J}	"	"	"	38	"	"	"	"	"
H27	Q_{2K}	"	"	"	17	"	"	"	"	"
H28	Q_{2L}	"	"	"	37	"	"	"	"	"
H29	Q_{2M}	"	"	"	20	"	"	"	"	"
H30	Q_{2N}	"	"	"	40	"	"	"	"	"
H31	Q_{2O}	"	"	"	19	"	"	"	"	"
H32	Q_{2P}	"	"	"	39	"	"	"	"	"

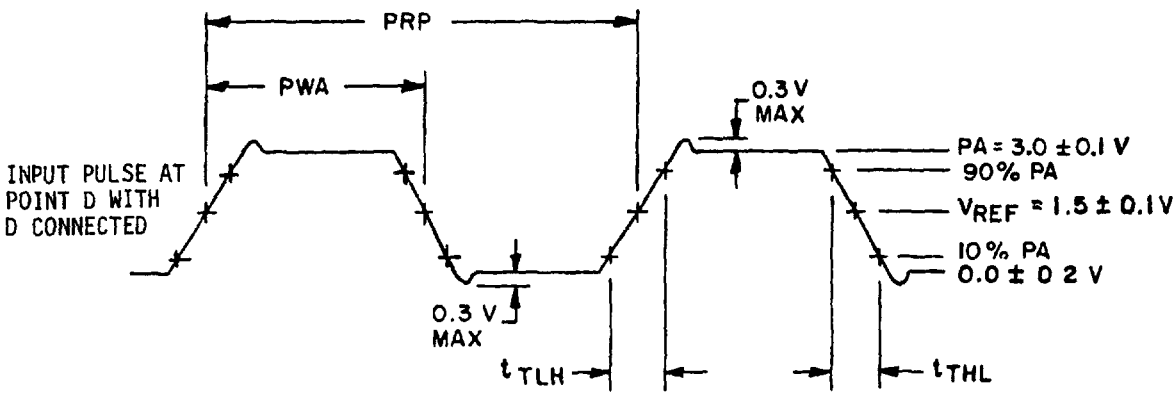
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NOTES:
1. R1 — R3 = 51.1Ω ±5%, 1/2 W.
2. Load each output with a load as specified on figure 16.
3. Allow a minimum of 17 clock pulses before making measurements for test conditions H1 - H32, then disable clock and make the measurements.

* FIGURE 14. Output voltage (low-level and high-level) test circuit.

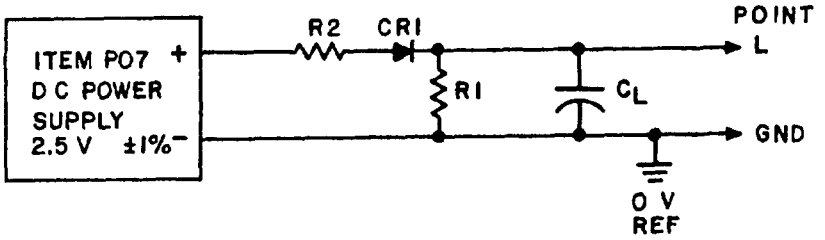
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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 50 \pm 25$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. All voltages are referenced to ground.

* FIGURE 15. Output voltage (low-level and high-level) test circuit waveform for tables IX and X.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 6.04 kΩ ±1%, 1/8 W.
3. R2 = Select a value for a current flow of 8 mA ±5% out of point L with point L held at 0.4 V dc.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions M1-M32, N1-N32, P1-P32, Q1-Q32, R1-R32, S1-S32, T1-T32, and U1-U32.
5. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions M33-M64, N33-N64, P33-P64, Q33-Q64, and R33-R64.
6. Capacitor C_L is not required for dc electrical test conditions G1-G32 and H1-H32.
7. Lead length for load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

* FIGURE 16. SJS logic load circuit.

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TABLE XI. Input current (high-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	A1	Input current, high level	4	5,8,11,36,16	0	80	µA	17
J2	CLEAR 1	"	5	4,8,11,36,16	-786	-1,008	"	"
J3	CLOCK 1	"	8	4,5,11,36,16	0	80	"	"
J4	A2	"	11	4,5,8,36,16	0	80	"	"
J5	CLEAR 2	"	36	4,5,8,11,16	-786	-1,008	"	"
J6	CLOCK 2	"	16	4,5,8,11,36	0	80	"	"

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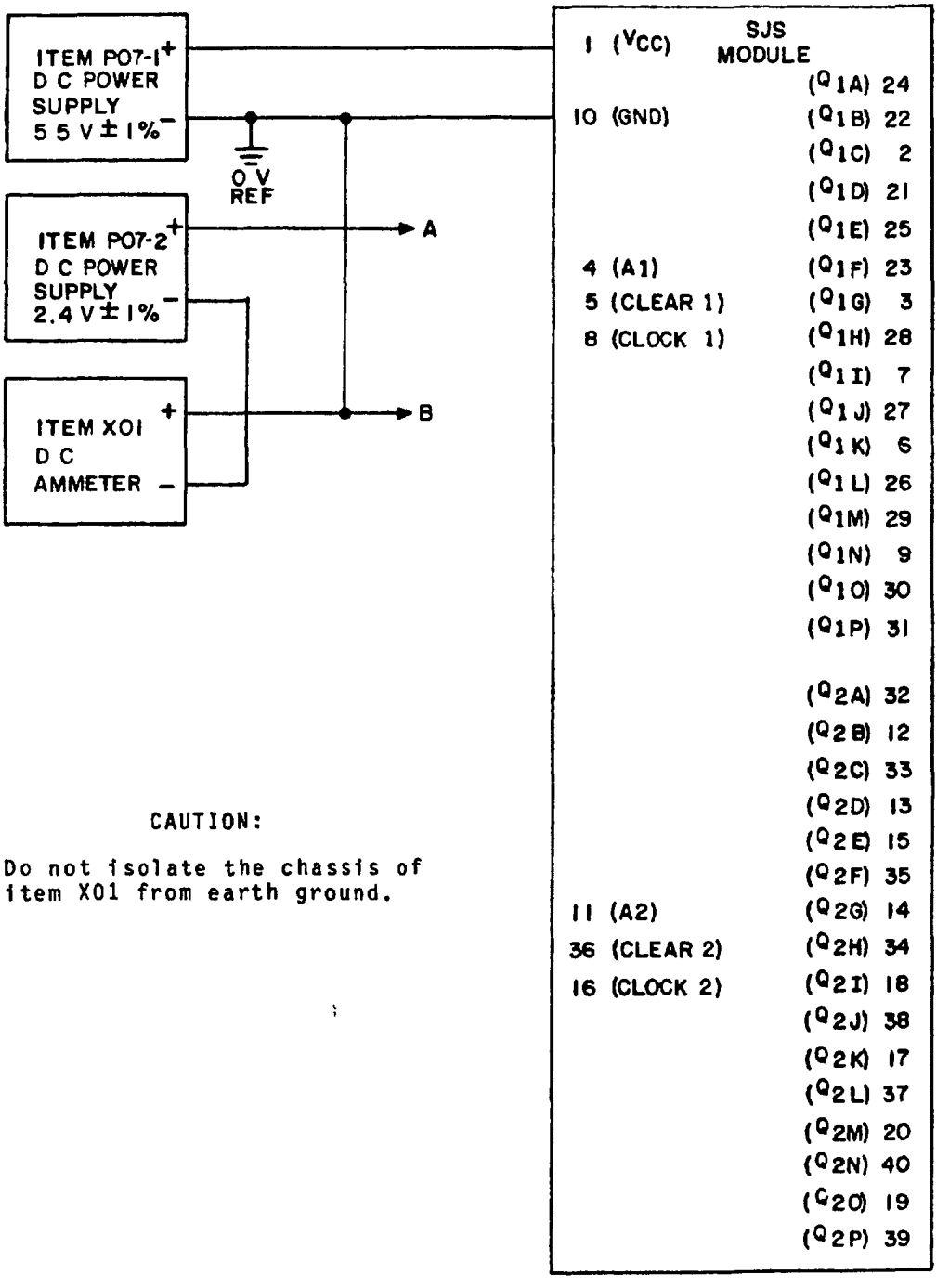


FIGURE 17. Input current (high-level) test circuit for table XI.

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*TABLE XII. Input current (low-level) test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
K1	A1	Input current, low level	4	-0.01	-1.73	mA	18
K2	CLEAR 1	"	5	-1.56	-5.21	"	"
K3	CLOCK 1	"	8	-0.01	-3.45	"	"
K4	A2	"	11	-0.01	-1.73	"	"
K5	CLEAR 2	"	36	-1.56	-5.21	"	"
K6	CLOCK 2	"	16	-0.01	-3.45	"	"

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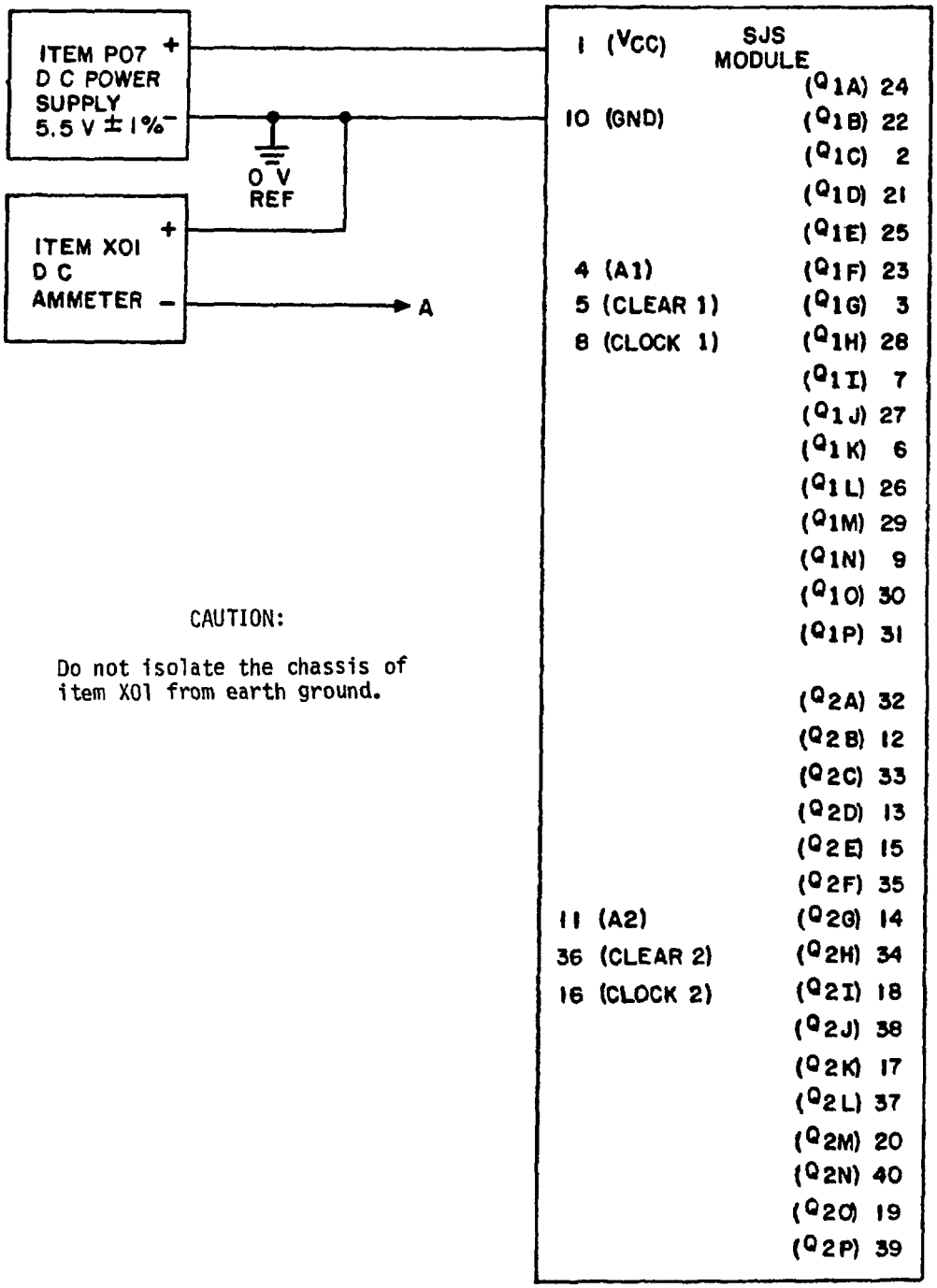


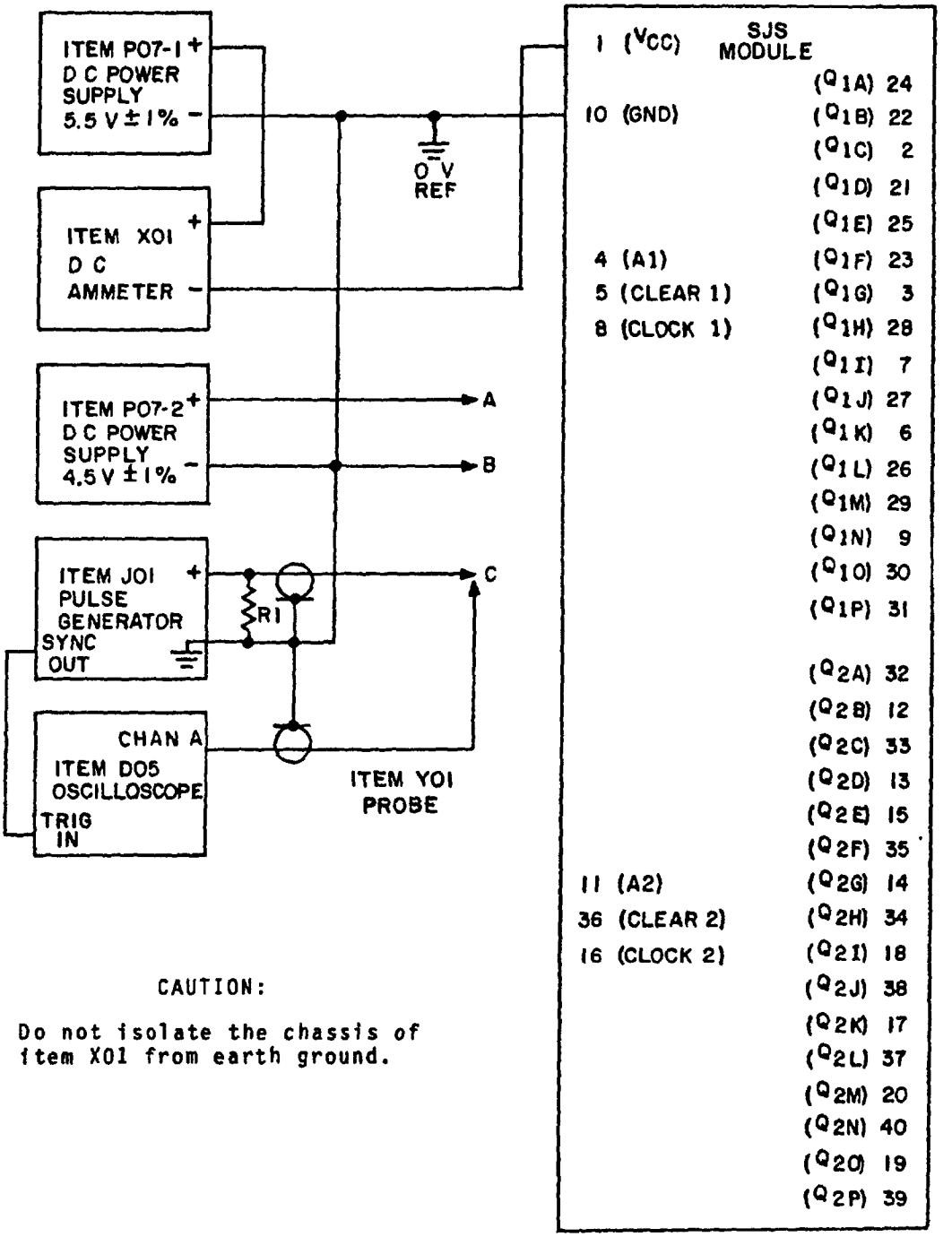
FIGURE 18. Input current (low-level) test circuit for table XII.

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* TABLE XIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Power supply current, low level	4, 8, 11, 16	5, 36	Open	50	216	mA	19, 20
L2	Power supply current, high level	4, 5, 11, 36	Open	8, 16	50	216	mA	19, 20

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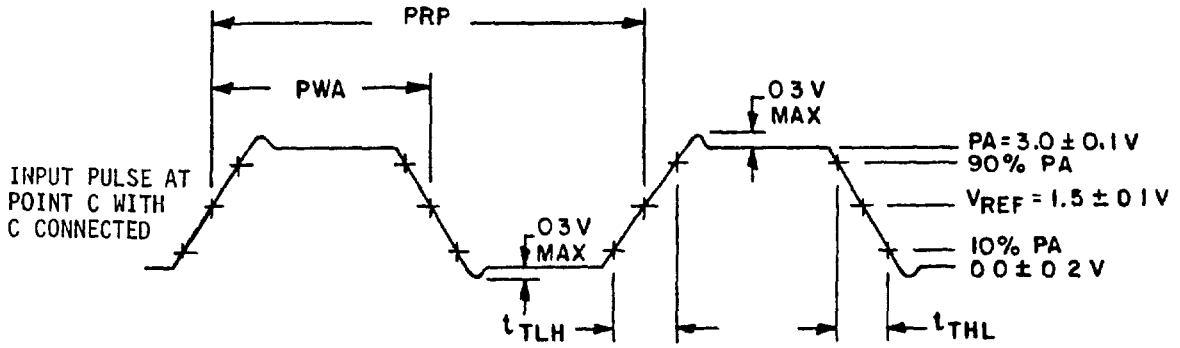
CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. Allow a minimum of 17 clock pulses before making measurements for test condition L2, then disable the clock and make the measurements.

* FIGURE 19. Power supply current (low-level and high-level) test circuit for table XIII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 50 \pm 25$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. All voltages are referenced to ground.

* FIGURE 20. Power supply current (low-level and high-level) test circuit waveform for table XIII.

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TABLE XIV. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	Q_{1A}	Transition time (t_{TLH}), C_L = 50 pF	5	8	4	24	5	85	ns	21,22
M2	Q_{1B}	"	"	"	"	22	"	"	"	"
M3	Q_{1C}	"	"	"	"	2	"	"	"	"
M4	Q_{1D}	"	"	"	"	21	"	"	"	"
M5	Q_{1E}	"	"	"	"	25	"	"	"	"
M6	Q_{1F}	"	"	"	"	23	"	"	"	"
M7	Q_{1G}	"	"	"	"	3	"	"	"	"
M8	Q_{1H}	"	"	"	"	28	"	"	"	"
M9	Q_{1I}	"	"	"	"	7	"	"	"	"
M10	Q_{1J}	"	"	"	"	27	"	"	"	"
M11	Q_{1K}	"	"	"	"	6	"	"	"	"
M12	Q_{1L}	"	"	"	"	26	"	"	"	"
M13	Q_{1M}	"	"	"	"	29	"	"	"	"
M14	Q_{1N}	"	"	"	"	9	"	"	"	"
M15	Q_{1O}	"	"	"	"	30	"	"	"	"
M16	Q_{1P}	"	"	"	"	31	"	"	"	"

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TABLE XIV. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M17	Q_{2A}	Transition time (t_{TLH}), C_L = 50 pF	36	16	11	32	5	85	ns	21,22
M18	Q_{2B}	"	"	"	"	12	"	"	"	"
M19	Q_{2C}	"	"	"	"	33	"	"	"	"
M20	Q_{2D}	"	"	"	"	13	"	"	"	"
M21	Q_{2E}	"	"	"	"	15	"	"	"	"
M22	Q_{2F}	"	"	"	"	35	"	"	"	"
M23	Q_{2G}	"	"	"	"	14	"	"	"	"
M24	Q_{2H}	"	"	"	"	34	"	"	"	"
M25	Q_{2I}	"	"	"	"	18	"	"	"	"
M26	Q_{2J}	"	"	"	'	38	"	"	"	"
M27	Q_{2K}	"	"	"	"	17	"	"	"	"
M28	Q_{2L}	"	"	"	"	37	"	"	"	"
M29	Q_{2M}	"	"	"	"	20	"	"	"	"
M30	Q_{2N}	"	"	"	"	40	"	"	"	"
M31	Q_{2O}	"	"	"	"	19	"	"	"	"
M32	Q_{2P}	"	"	"	"	39	"	"	"	"

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TABLE XIV. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M33	Q_{1A}	Transition time (t_{TLH}), C_L = 150 pF	5	8	4	24	15	130	ns	21,22
M34	Q_{1B}	"	"	"	"	22	"	"	"	"
M35	Q_{1C}	"	"	"	"	2	"	"	"	"
M36	Q_{1D}	"	"	"	"	21	"	"	"	"
M37	Q_{1E}	"	"	"	"	25	"	"	"	"
M38	Q_{1F}	"	"	"	"	23	"	"	"	"
M39	Q_{1G}	"	"	"	"	3	"	"	"	"
M40	Q_{1H}	"	"	"	"	28	"	"	"	"
M41	Q_{1I}	"	"	"	"	7	"	"	"	"
M42	Q_{1J}	"	"	"	"	27	"	"	"	"
M43	Q_{1K}	"	"	"	"	6	"	"	"	"
M44	Q_{1L}	"	"	"	"	26	"	"	"	"
M45	Q_{1M}	"	"	"	"	29	"	"	"	"
M46	Q_{1N}	"	"	"	"	9	"	"	"	"
M47	Q_{1O}	"	"	"	"	30	"	"	"	"
M48	Q_{1P}	"	"	"	"	31	"	"	"	"

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TABLE XIV. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M49	Q_{2A} Transition time (t_{TLH}), C_L = 150 pF	36	16	11	32	15	130	ns	21,22
M50	Q_{2B} "	"	"	"	12	"	"	"	"
M51	Q_{2C} "	"	"	"	33	"	"	"	"
M52	Q_{2D} "	"	"	"	13	"	"	"	"
M53	Q_{2E} "	"	"	"	15	"	"	"	"
M54	Q_{2F} "	"	"	"	35	"	"	"	"
M55	Q_{2G} "	"	"	"	14	"	"	"	"
M56	Q_{2H} "	"	"	"	34	"	"	"	"
M57	Q_{2I} "	"	"	"	18	"	"	"	"
M58	Q_{2J} "	"	"	"	38	"	"	"	"
M59	Q_{2K} "	"	"	"	17	"	"	"	"
M60	Q_{2L} "	"	"	"	37	"	"	"	"
M61	Q_{2M} "	"	"	"	20	"	"	"	"
M62	Q_{2N} "	"	"	"	40	"	"	"	"
M63	Q_{2O} "	"	"	"	19	"	"	"	"
M64	Q_{2P} "	"	"	"	39	"	"	"	"

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TABLE XV. Transition time (t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Q_{1A}	Transition time (t_{THL}), C_L = 50 pF	5	8	4	24	4	20	ns	21,22
N2	Q_{1B}	"	"	"	"	22	"	"	"	"
N3	Q_{1C}	"	"	"	"	2	"	"	"	"
N4	Q_{1D}	"	"	"	"	21	"	"	"	"
N5	Q_{1E}	"	"	"	"	25	"	"	"	"
N6	Q_{1F}	"	"	"	"	23	"	"	"	"
N7	Q_{1G}	"	"	"	"	3	"	"	"	"
N8	Q_{1H}	"	"	"	"	28	"	"	"	"
N9	Q_{1I}	"	"	"	"	7	"	"	"	"
N10	Q_{1J}	"	"	"	"	27	"	"	"	"
N11	Q_{1K}	"	"	"	"	6	"	"	"	"
N12	Q_{1L}	"	"	"	"	26	"	"	"	"
N13	Q_{1M}	"	"	"	"	29	"	"	"	"
N14	Q_{1N}	"	"	"	"	9	"	"	"	"
N15	Q_{1O}	"	"	"	"	30	"	"	"	"
N16	Q_{1P}	"	"	"	"	31	"	"	"	"

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TABLE XV. Transition time (t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N17	Q_{2A}	Transition time (t_{THL}), C_L = 50 pF	36	16	11	32	4	20	ns	21,22
N18	Q_{2B}	"	"	"	"	12	"	"	"	"
N19	Q_{2C}	"	"	"	"	33	"	"	"	"
N20	Q_{2D}	"	"	"	"	13	"	"	"	"
N21	Q_{2E}	"	"	"	"	15	"	"	"	"
N22	Q_{2F}	"	"	"	"	35	"	"	"	"
N23	Q_{2G}	"	"	"	"	14	"	"	"	"
N24	Q_{2H}	"	"	"	"	34	"	"	"	"
N25	Q_{2I}	"	"	"	"	18	"	"	"	"
N26	Q_{2J}	"	"	"	"	38	"	"	"	"
N27	Q_{2K}	"	"	"	"	17	"	"	"	"
N28	Q_{2L}	"	"	"	"	37	"	"	"	"
N29	Q_{2M}	"	"	"	"	20	"	"	"	"
N30	Q_{2N}	"	"	"	"	40	"	"	"	"
N31	Q_{2O}	"	"	"	"	19	"	"	"	"
N32	Q_{2P}	"	"	"	"	39	"	"	"	"

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TABLE XV. Transition time (t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N33	Q_{1A}	Transition time (t_{THL}), C_L = 150 pF	5	8	4	24	5	30	ns	21,22
N34	Q_{1B}	"	"	"	"	22	"	"	"	"
N35	Q_{1C}	"	"	"	"	2	"	"	"	"
N36	Q_{1D}	"	"	"	"	21	"	"	"	"
N37	Q_{1E}	"	"	"	"	25	"	"	"	"
N38	Q_{1F}	"	"	"	"	23	"	"	"	"
N39	Q_{1G}	"	"	"	"	3	"	"	"	"
N40	Q_{1H}	"	"	"	"	28	"	"	"	"
N41	Q_{1I}	"	"	"	"	7	"	"	"	"
N42	Q_{1J}	"	"	"	"	27	"	"	"	"
N43	Q_{1K}	"	"	"	"	6	"	"	"	"
N44	Q_{1L}	"	"	"	"	26	"	"	"	"
N45	Q_{1M}	"	"	"	"	29	"	"	"	"
N46	Q_{1N}	"	"	"	"	9	"	"	"	"
N47	Q_{1O}	"	"	"	"	30	"	"	"	"
N48	Q_{1P}	"	"	"	"	31	"	"	"	"

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TABLE XV. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N49	Q_{2A} Transition time (t_{THL}), C_L = 150 pF	36	16	11	32	5	30	ns	21,22
N50	Q_{2B} "	"	"	"	12	"	"	"	"
N51	Q_{2C} "	"	"	"	33	"	"	"	"
N52	Q_{2D} "	"	"	"	13	"	"	"	"
N53	Q_{2E} "	"	"	"	15	"	"	"	"
N54	Q_{2F} "	"	"	"	35	"	"	"	"
N55	Q_{2G} "	"	"	"	14	"	"	"	"
N56	Q_{2H} "	"	"	"	34	"	"	"	"
N57	Q_{2I} "	"	"	"	18	"	"	"	"
N58	Q_{2J} "	"	"	"	38	"	"	"	"
N59	Q_{2K} "	"	"	"	17	"	"	"	"
N60	Q_{2L} "	"	"	"	37	"	"	"	"
N61	Q_{2M} "	"	"	"	20	"	"	"	"
N62	Q_{2N} "	"	"	"	40	"	"	"	"
N63	Q_{2O} "	"	"	"	19	"	"	"	"
N64	Q_{2P} "	"	"	"	39	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	Q_{1A}	Propagation delay time (clock to output) (t_{PLH}) C_L = 50 pF	5	8	4	24	10	30	ns	21,22
P2	Q_{1B}	"	"	"	"	22	"	"	"	"
P3	Q_{1C}	"	"	"	"	2	"	"	"	"
P4	Q_{1D}	"	"	"	"	21	"	"	"	"
P5	Q_{1E}	"	"	"	"	25	"	"	"	"
P6	Q_{1F}	"	"	"	"	23	"	"	"	"
P7	Q_{1G}	"	"	"	"	3	"	"	"	"
P8	Q_{1H}	"	"	"	"	28	"	"	"	"
P9	Q_{1I}	"	"	"	"	7	"	"	"	"
P10	Q_{1J}	"	"	"	"	27	"	"	"	"
P11	Q_{1K}	"	"	"	"	6	"	"	"	"
P12	Q_{1L}	"	"	"	"	26	"	"	"	"
P13	Q_{1M}	"	"	"	"	29	"	"	"	"
P14	Q_{1N}	"	"	"	"	9	"	"	"	"
P15	Q_{1O}	"	"	"	"	30	"	"	"	"
P16	Q_{1P}	"	"	"	"	31	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P17	Q_{2A}	Propagation delay time (clock to output) (t_{PLH}) C_L = 50 pF	36	16	11	32	10	30	ns	21,22
P18	Q_{2B}	"	"	"	"	12	"	"	"	"
P19	Q_{2C}	"	"	"	"	33	"	"	"	"
P20	Q_{2D}	"	"	"	"	13	"	"	"	"
P21	Q_{2E}	"	"	"	"	15	"	"	"	"
P22	Q_{2F}	"	"	"	"	35	"	"	"	"
P23	Q_{2G}	"	"	"	"	14	"	"	"	"
P24	Q_{2H}	"	"	"	"	34	"	"	"	"
P25	Q_{2I}	"	"	"	"	18	"	"	"	"
P26	Q_{2J}	"	"	"	"	38	"	"	"	"
P27	Q_{2K}	"	"	"	"	17	"	"	"	"
P28	Q_{2L}	"	"	"	"	37	"	"	"	"
P29	Q_{2M}	"	"	"	"	20	"	"	"	"
P30	Q_{2N}	"	"	"	"	40	"	"	"	"
P31	Q_{2O}	"	"	"	"	19	"	"	"	"
P32	Q_{2P}	"	"	"	"	39	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P33	Q_{1A}	Propagation delay time (clock to output) (t_{PLH}) C_L = 150 pF	5	8	4	24	10	45	ns	21,22
P34	Q_{1B}	"	"	"	"	22	"	"	"	"
P35	Q_{1C}	"	"	"	"	2	"	"	"	"
P36	Q_{1D}	"	"	"	"	21	"	"	"	"
P37	Q_{1E}	"	"	"	"	25	"	"	"	"
P38	Q_{1F}	"	"	"	"	23	"	"	"	"
P39	Q_{1G}	"	"	"	"	3	"	"	"	"
P40	Q_{1H}	"	"	"	"	28	"	"	"	"
P41	Q_{1I}	"	"	"	"	7	"	"	"	"
P42	Q_{1J}	"	"	"	"	27	"	"	"	"
P43	Q_{1K}	"	"	"	"	6	"	"	"	"
P44	Q_{1L}	"	"	"	"	26	"	"	"	"
P45	Q_{1M}	"	"	"	"	29	"	"	"	"
P46	Q_{1N}	"	"	"	"	9	"	"	"	"
P47	Q_{1O}	"	"	"	"	30	"	"	"	"
P48	Q_{1P}	"	"	"	"	31	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P49	Q_{2A}	Propagation delay time (clock to output) (t_{PLH}) C_L = 150 pF	36	16	11	32	10	45	ns	21,22
P50	Q_{2B}	"	"	"	"	12	"	"	"	"
P51	Q_{2C}	"	"	"	"	33	"	"	"	"
P52	Q_{2D}	"	"	"	"	13	"	"	"	"
P53	Q_{2E}	"	"	"	"	15	"	"	"	"
P54	Q_{2F}	"	"	"	"	35	"	"	"	"
P55	Q_{2G}	"	"	"	"	14	"	"	"	"
P56	Q_{2H}	"	"	"	"	34	"	"	"	"
P57	Q_{2I}	"	"	"	"	18	"	"	"	"
P58	Q_{2J}	"	"	"	"	38	"	"	"	"
P59	Q_{2K}	"	"	"	"	17	"	"	"	"
P60	Q_{2L}	"	"	"	"	37	"	"	"	"
P61	Q_{2M}	"	"	"	"	20	"	"	"	"
P62	Q_{2N}	"	"	"	"	40	"	"	"	"
P63	Q_{2O}	"	"	"	"	19	"	"	"	"
P64	Q_{2P}	"	"	"	"	39	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	Q_{1A}	Propagation delay time (clock to output) (t_{PHL}) C_L = 50 pF	5	8	4	24	10	37	ns	21,22
Q2	Q_{1B}	"	"	"	"	22	"	"	"	"
Q3	Q_{1C}	"	"	"	"	2	"	"	"	"
Q4	Q_{1D}	"	"	"	"	21	"	"	"	"
Q5	Q_{1E}	"	"	"	"	25	"	"	"	"
Q6	Q_{1F}	"	"	"	"	23	"	"	"	"
Q7	Q_{1G}	"	"	"	"	3	"	"	"	"
Q8	Q_{1H}	"	"	"	"	28	"	"	"	"
Q9	Q_{1I}	"	"	"	"	7	"	"	"	"
Q10	Q_{1J}	"	"	"	"	27	"	"	"	"
Q11	Q_{1K}	"	"	"	"	6	"	"	"	"
Q12	Q_{1L}	"	"	"	"	26	"	"	"	"
Q13	Q_{1M}	"	"	"	"	29	"	"	"	"
Q14	Q_{1N}	"	"	"	"	9	"	"	"	"
Q15	Q_{1O}	"	"	"	"	30	"	"	"	"
Q16	Q_{1P}	"	"	"	"	31	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q17	Q_{2A}	Propagation delay time (clock to output) (t_{PHL}) C_L = 50 pF	36	16	11	32	10	37	ns	21,22
Q18	Q_{2B}	"	"	"	"	12	"	"	"	"
Q19	Q_{2C}	"	"	"	"	33	"	"	"	"
Q20	Q_{2D}	"	"	"	"	13	"	"	"	"
Q21	Q_{2E}	"	"	"	"	15	"	"	"	"
Q22	Q_{2F}	"	"	"	"	35	"	"	"	"
Q23	Q_{2G}	"	"	"	"	14	"	"	"	"
Q24	Q_{2H}	"	"	"	"	34	"	"	"	"
Q25	Q_{2I}	"	"	"	"	18	"	"	"	"
Q26	Q_{2J}	"	"	"	"	38	"	"	"	"
Q27	Q_{2K}	"	"	"	"	17	"	"	"	"
Q28	Q_{2L}	"	"	"	"	37	"	"	"	"
Q29	Q_{2M}	"	"	"	"	20	"	"	"	"
Q30	Q_{2N}	"	"	"	"	40	"	"	"	"
Q31	Q_{2O}	"	"	"	"	19	"	"	"	"
Q32	Q_{2P}	"	"	"	"	39	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q33	Q_{1A}	Propagation delay time (clock to output) (t_{PHL}) C_L = 150 pF	5	8	4	24	10	47	ns	21,22
Q34	Q_{1B}	"	"	"	"	22	"	"	"	"
Q35	Q_{1C}	"	"	"	"	2	"	"	"	"
Q36	Q_{1D}	"	"	"	"	21	"	"	"	"
Q37	Q_{1E}	"	"	"	"	25	"	"	"	"
Q38	Q_{1F}	"	"	"	"	23	"	"	"	"
Q39	Q_{1G}	"	"	"	"	3	"	"	"	"
Q40	Q_{1H}	"	"	"	"	28	"	"	"	"
Q41	Q_{1I}	"	"	"	"	7	"	"	"	"
Q42	Q_{1J}	"	"	"	"	27	"	"	"	"
Q43	Q_{1K}	"	"	"	"	6	"	"	"	"
Q44	Q_{1L}	"	"	"	"	26	"	"	"	"
Q45	Q_{1M}	"	"	"	"	29	"	"	"	"
Q46	Q_{1N}	"	"	"	"	9	"	"	"	"
Q47	Q_{1O}	"	"	"	"	30	"	"	"	"
Q48	Q_{1P}	"	"	"	"	31	"	"	"	"

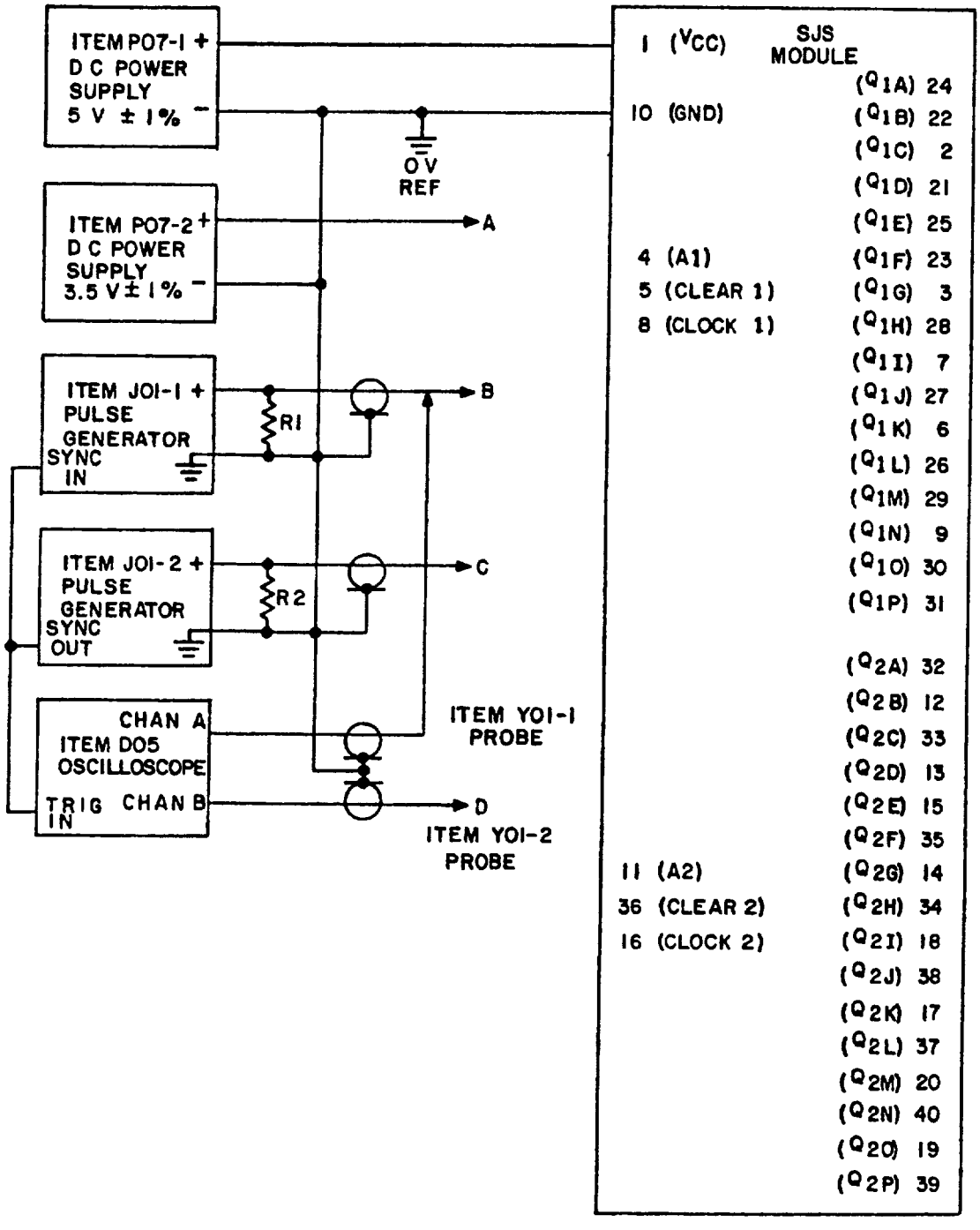
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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q49	Q_{2A}	Propagation delay time (clock to output) (t_{PHL}) C_L = 150 pF	36	16	11	32	10	47	ns	21,22
Q50	Q_{2B}	"	"	"	"	12	"	"	"	"
Q51	Q_{2C}	"	"	"	"	33	"	"	"	"
Q52	Q_{2D}	"	"	"	"	13	"	"	"	"
Q53	Q_{2E}	"	"	"	"	15	"	"	"	"
Q54	Q_{2F}	"	"	"	"	35	"	"	"	"
Q55	Q_{2G}	"	"	"	"	14	"	"	"	"
Q56	Q_{2H}	"	"	"	"	34	"	"	"	"
Q57	Q_{2I}	"	"	"	"	18	"	"	"	"
Q58	Q_{2J}	"	"	"	"	38	"	"	"	"
Q59	Q_{2K}	"	"	"	"	17	"	"	"	"
Q60	Q_{2L}	"	"	"	"	37	"	"	"	"
Q61	Q_{2M}	"	"	"	"	20	"	"	"	"
Q62	Q_{2N}	"	"	"	"	40	"	"	"	"
Q63	Q_{2O}	"	"	"	"	19	"	"	"	"
Q64	Q_{2P}	"	"	"	"	39	"	"	"	"

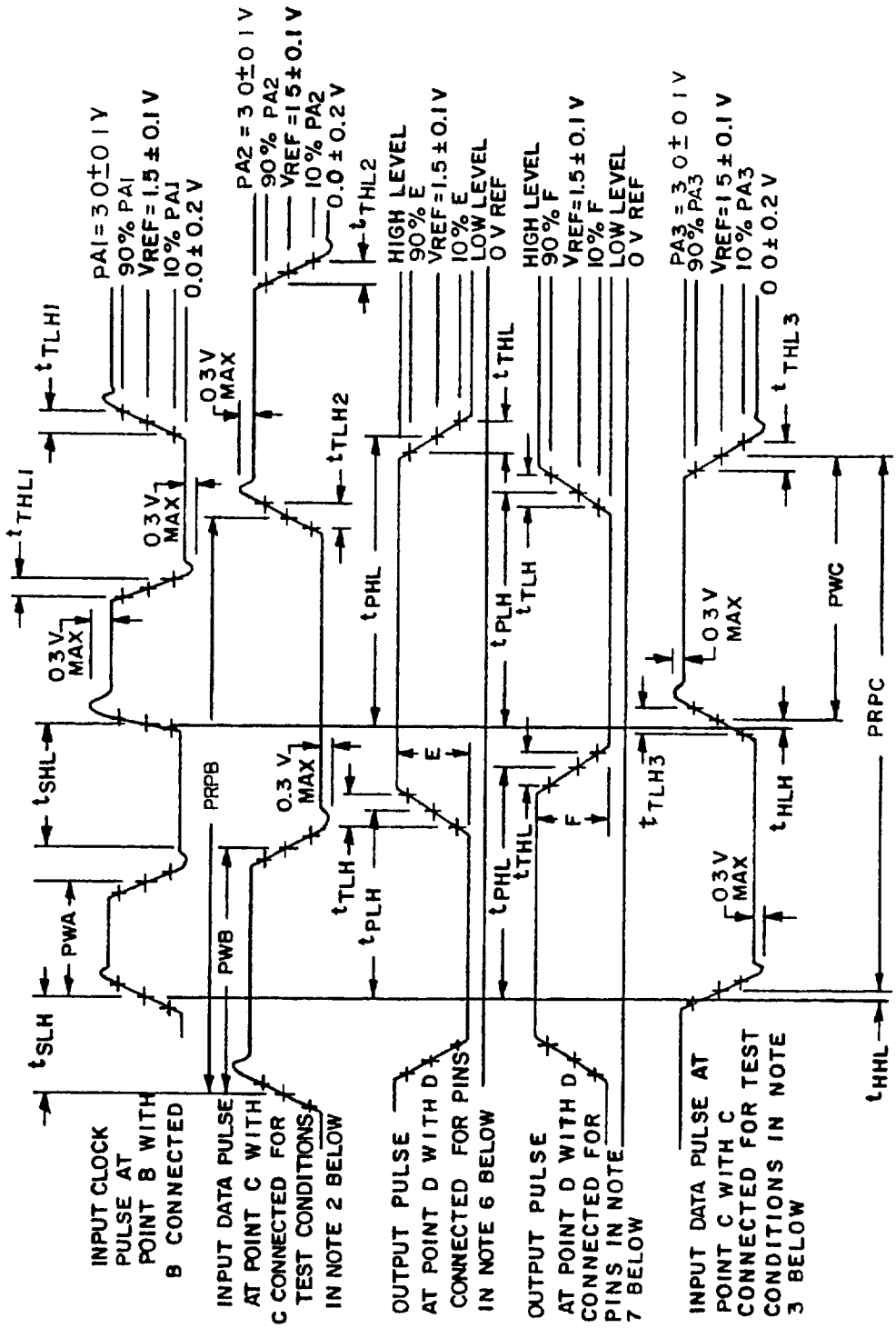
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. Load each output with load as specified on figure 16.

FIGURE 21. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIV, XV, XVI, and XVII.

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* FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIV, XV, XVI, and XVII.

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NOTES:
1. Input pulse parameters for item J01-1:
   Put item J01-1 in the double pulse mode.
   $t_{TLH1} = t_{THL1} = 10 \pm 1$ ns.
   PWA = 100 ±10 ns.
2. Input pulse parameters for item J01-2 for test conditions M1-M64 and N1-N64:
   $t_{TLH2} = t_{THL2} = 10 \pm 1$ ns.
   PWB = 200 ±20 ns.
   PRPB = 400 ±40 ns.
3. Input pulse parameters for item J01-2 for test conditions P1-P64 and Q1-Q64:
   $t_{TLH3} = t_{THL3} = 10 \pm 1$ ns.
   PWC = 200 ±4 ns.
   PRPC = 400 ±6 ns.
4. $t_{SLH} = t_{SHL} = 15$ ns ±5%.
5. $t_{HLH} = t_{HHL} = 5 +1, -0$ ns.
6. Pins 24, 2, 25, 3, 7, 6, 29, 30, 32, 33, 15, 14, 18, 17, 20, and 19.
7. Pins 22, 21, 23, 28, 27, 26, 9, 31, 12, 13, 35, 34, 38, 37, 40, and 39.
8. Verify that all outputs have a high level $\geq$ 2.4 V and a low level $\leq$ 0.4 V before making any measurements.
9. The physical length of electrical conductors from the resistors R1 and R2 to input pins, from items Y01-1 and Y01-2 to input pins, and from load circuits to output pins shall be less than 1 inch (25.4 mm).
10. All voltages are referenced to ground.

* FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIV, XV, XVI, and XVII - Continued.

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TABLE XVIII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	Q_{1A}	Propagation delay time (clear to output) (t_{PHL}) C_L = 50 pF	4	8	5	24	10	42	ns	23,24
R2	Q_{1B}	"	"	"	"	22	"	"	"	"
R3	Q_{1C}	"	"	"	"	2	"	"	"	"
R4	Q_{1D}	"	"	"	"	21	"	"	"	"
R5	Q_{1E}	"	"	"	"	25	"	"	"	"
R6	Q_{1F}	"	"	"	"	23	"	"	"	"
R7	Q_{1G}	"	"	"	"	3	"	"	"	"
R8	Q_{1H}	"	"	"	"	28	"	"	"	"
R9	Q_{1I}	"	"	"	"	7	"	"	"	"
R10	Q_{1J}	"	"	"	"	27	"	"	"	"
R11	Q_{1K}	"	"	"	"	6	"	"	"	"
R12	Q_{1L}	"	"	"	"	26	"	"	"	"
R13	Q_{1M}	"	"	"	"	29	"	"	"	"
R14	Q_{1N}	"	"	"	"	9	"	"	"	"
R15	Q_{1O}	"	"	"	"	30	"	"	"	"
R16	Q_{1P}	"	"	"	"	31	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R17	Q_{2A}	Propagation delay time (clear to output) (t_{PHL}) C_L = 50 pF	11	16	36	32	10	42	ns	23,24
R18	Q_{2B}	"	"	"	"	12	"	"	"	"
R19	Q_{2C}	"	"	"	"	33	"	"	"	"
R20	Q_{2D}	"	"	"	"	13	"	"	"	"
R21	Q_{2E}	"	"	"	"	15	"	"	"	"
R22	Q_{2F}	"	"	"	"	35	"	"	"	"
R23	Q_{2G}	"	"	"	"	14	"	"	"	"
R24	Q_{2H}	"	"	"	"	34	"	"	"	"
R25	Q_{2I}	"	"	"	"	18	"	"	"	"
R26	Q_{2J}	"	"	"	"	38	"	"	"	"
R27	Q_{2K}	"	"	"	"	17	"	"	"	"
R28	Q_{2L}	"	"	"	"	37	"	"	"	"
R29	Q_{2M}	"	"	"	"	20	"	"	"	"
R30	Q_{2N}	"	"	"	"	40	"	"	"	"
R31	Q_{2O}	"	"	"	"	19	"	"	"	"
R32	Q_{2P}	"	"	"	"	39	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

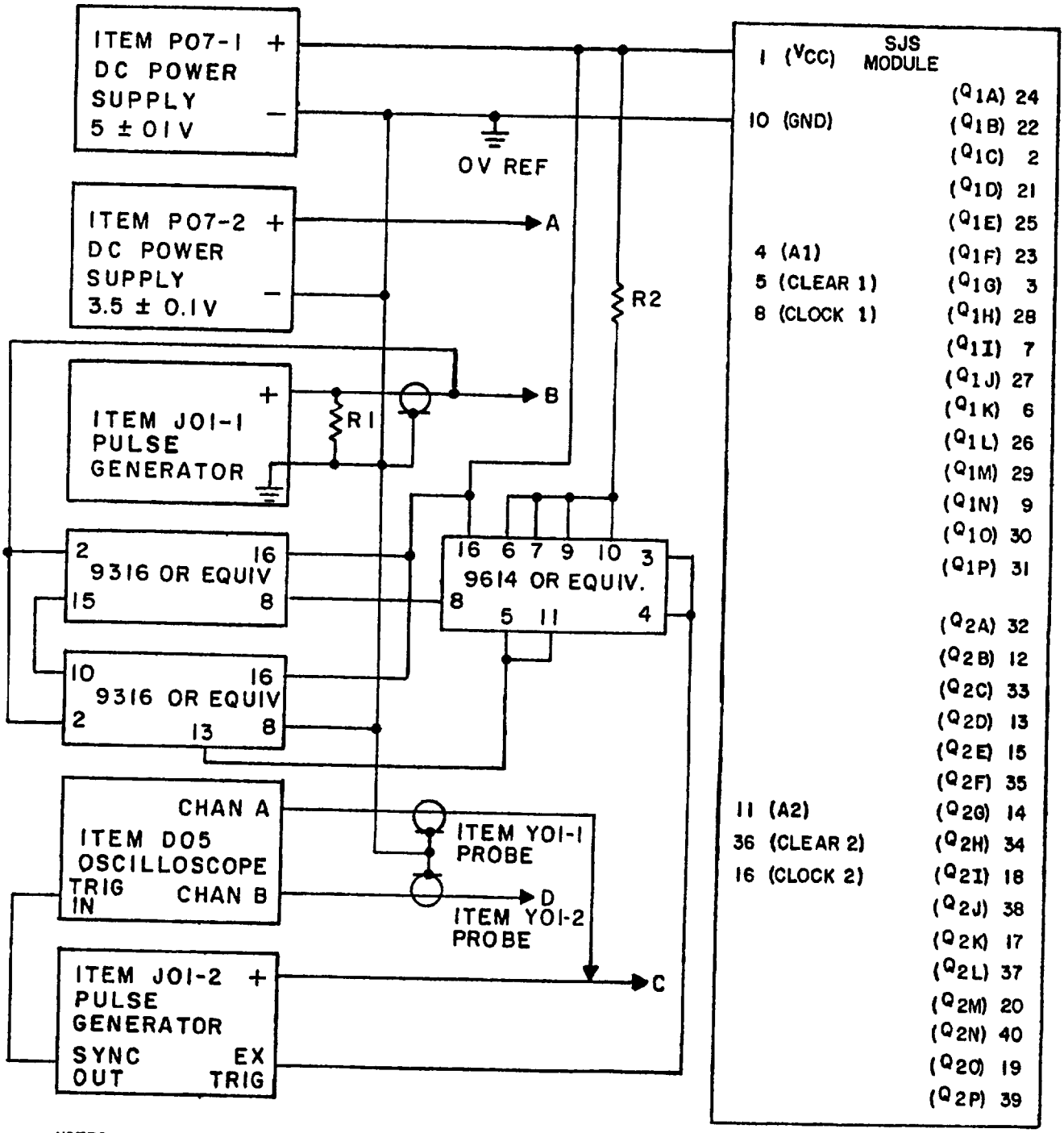
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R33	Q_{1A}	Propagation delay time (clear to output) (t_{PHL}) C_L = 150 pF	4	8	5	24	10	52	ns	23,24
R34	Q_{1B}	"	"	"	"	22	"	"	"	"
R35	Q_{1C}	"	"	"	"	2	"	"	"	"
R36	Q_{1D}	"	"	"	"	21	"	"	"	"
R37	Q_{1E}	"	"	"	"	25	"	"	"	"
R38	Q_{1F}	"	"	"	"	23	"	"	"	"
R39	Q_{1G}	"	"	"	"	3	"	"	"	"
R40	Q_{1H}	"	"	"	"	28	"	"	"	"
R41	Q_{1I}	"	"	"	"	7	"	"	"	"
R42	Q_{1J}	"	"	"	"	27	"	"	"	"
R43	Q_{1K}	"	"	"	"	6	"	"	"	"
R44	Q_{1L}	"	"	"	"	26	"	"	"	"
R45	Q_{1M}	"	"	"	"	29	"	"	"	"
R46	Q_{1N}	"	"	"	"	9	"	"	"	"
R47	Q_{1O}	"	"	"	"	30	"	"	"	"
R48	Q_{1P}	"	"	"	"	31	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R49	Q_{2A}	Propagation delay time (clear to output) (t_{PHL}) C_L = 150 pF	11	16	36	32	10	52	ns	23,24
R50	Q_{2B}	"	"	"	"	12	"	"	"	"
R51	Q_{2C}	"	"	"	"	33	"	"	"	"
R52	Q_{2D}	"	"	"	"	13	"	"	"	"
R53	Q_{2E}	"	"	"	"	15	"	"	"	"
R54	Q_{2F}	"	"	"	"	35	"	"	"	"
R55	Q_{2G}	"	"	"	"	14	"	"	"	"
R56	Q_{2H}	"	"	"	"	34	"	"	"	"
R57	Q_{2I}	"	"	"	"	18	"	"	"	"
R58	Q_{2J}	"	"	"	"	38	"	"	"	"
R59	Q_{2K}	"	"	"	"	17	"	"	"	"
R60	Q_{2L}	"	"	"	"	37	"	"	"	"
R61	Q_{2M}	"	"	"	"	20	"	"	"	"
R62	Q_{2N}	"	"	"	"	40	"	"	"	"
R63	Q_{2O}	"	"	"	"	19	"	"	"	"
R64	Q_{2P}	"	"	"	"	39	"	"	"	"

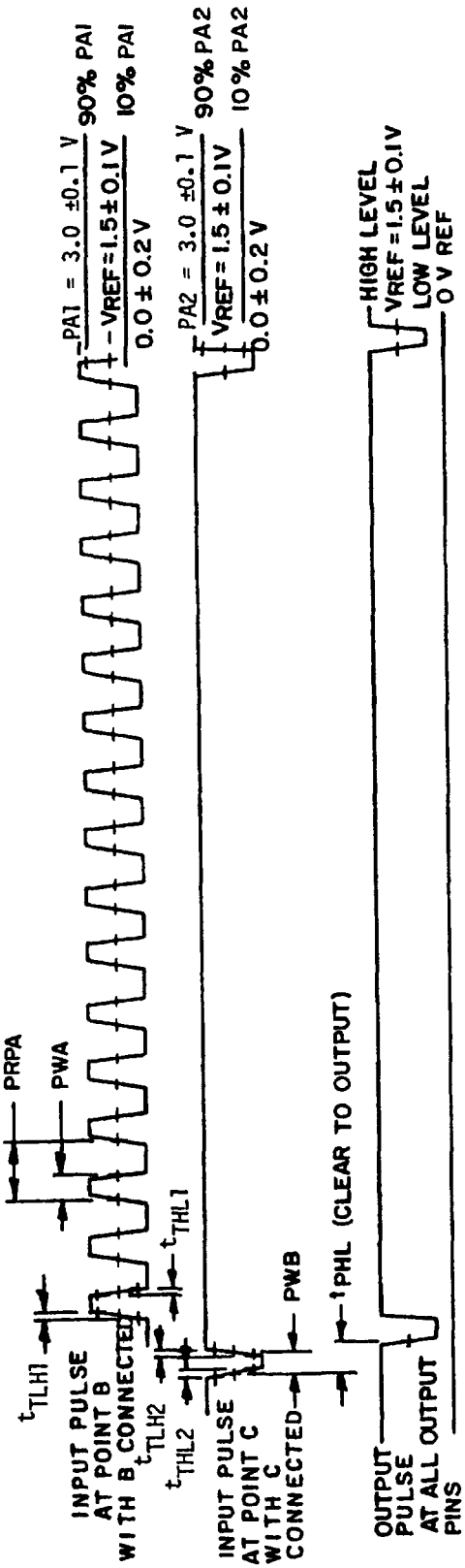
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R2 = 1 kΩ ±1%, 1/4 W.
3. Load each output with load as specified on figure 16.
4. Before making the measurements verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.5 V.
5. The lead length of electrical conductors between resistors R1 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).

* FIGURE 23. Propagation delay time (t_{PHL}) test circuit for table XVIII.

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NOTES:
1. Input pulse parameters for item J01-1:
   PWA = 200 ±20 ns.
   PRPA = 400 ±40 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   PA1 = 3.0 ±0.1 V.
2. Input pulse parameters for item J01-2:
   PWB = 20 ±2 ns.
   $t_{TLH2} = t_{THL2}$ = 4 ±1 ns.
   PA2 = 3.0 ±0.1 V.
3. All voltages are referenced to ground.

* FIGURE 24. Propagation delay time (t_{PHL}) test circuit waveforms for table XVIII.

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TABLE XIX. Maximum shift frequency test.

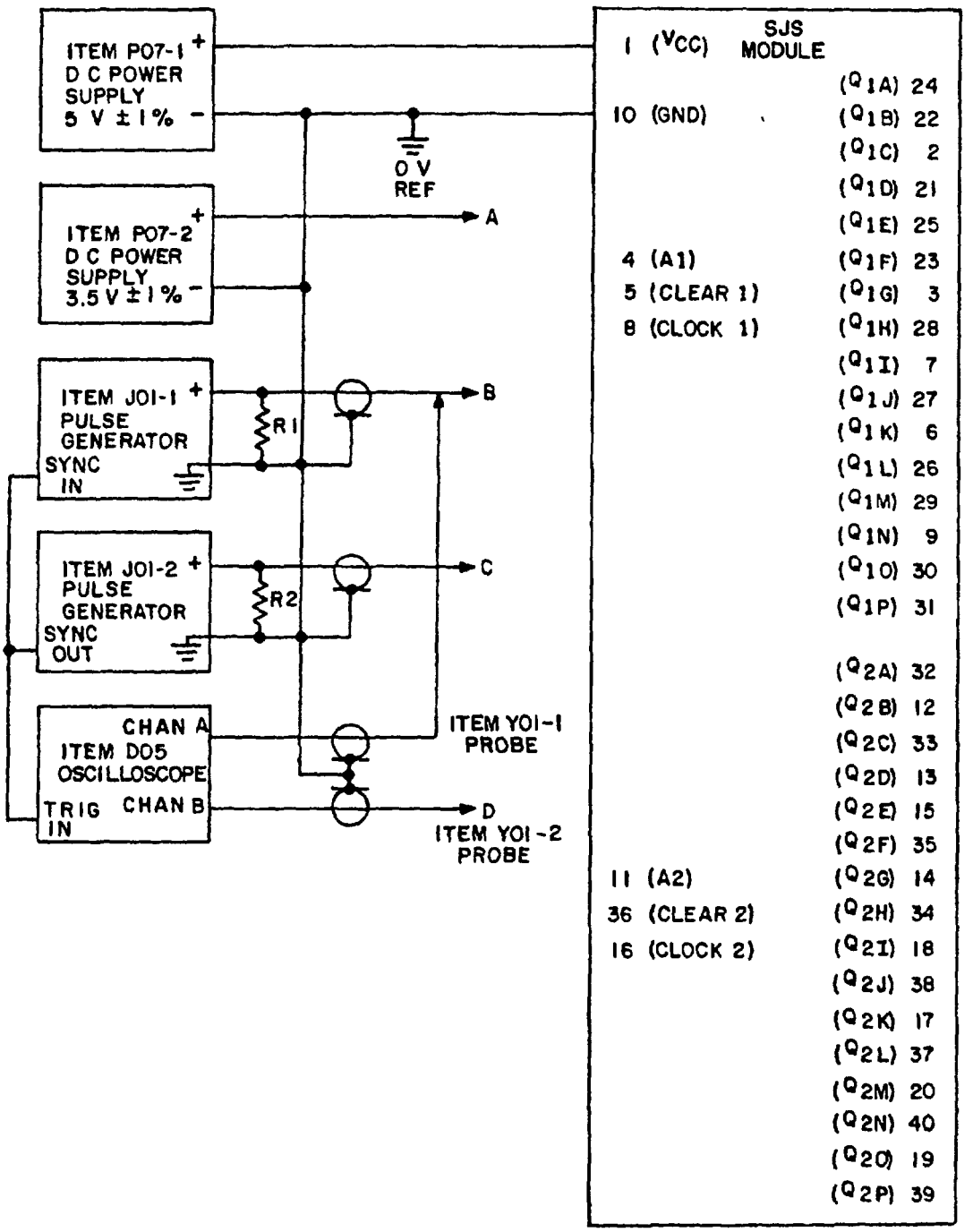
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	Q_{1A} Maximum shift frequency	5	8	4	24	72	88	ns	25,26
S2	Q_{1B} "	"	"	"	22	"	"	"	"
S3	Q_{1C} "	"	"	"	2	"	"	"	"
S4	Q_{1D} "	"	"	"	21	"	"	"	"
S5	Q_{1E} "	"	"	"	25	"	"	"	"
S6	Q_{1F} "	"	"	"	23	"	"	"	"
S7	Q_{1G} "	"	"	"	3	"	"	"	"
S8	Q_{1H} "	"	"	"	28	"	"	"	"
S9	Q_{1I} "	"	"	"	7	"	"	"	"
S10	Q_{1J} "	"	"	"	27	"	"	"	"
S11	Q_{1K} "	"	"	"	6	"	"	"	"
S12	Q_{1L} "	"	"	"	26	"	"	"	"
S13	Q_{1M} "	"	"	"	29	"	"	"	"
S14	Q_{1N} "	"	"	"	9	"	"	"	"
S15	Q_{1O} "	"	"	"	30	"	"	"	"
S16	Q_{1P} "	"	"	"	31	"	"	"	"

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TABLE XIX. Maximum shift frequency test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S17	Q_{2A}	Maximum shift frequency	36	16	11	32	72	88	ns	25,26
S18	Q_{2B}	"	"	"	"	12	"	"	"	"
S19	Q_{2C}	"	"	"	"	33	"	"	"	"
S20	Q_{2D}	"	"	"	"	13	"	"	"	"
S21	Q_{2E}	"	"	"	"	15	"	"	"	"
S22	Q_{2F}	"	"	"	"	35	"	"	"	"
S23	Q_{2G}	"	"	"	"	14	"	"	"	"
S24	Q_{2H}	"	"	"	"	34	"	"	"	"
S25	Q_{2I}	"	"	"	"	18	"	"	"	"
S26	Q_{2J}	"	"	"	"	38	"	"	"	"
S27	Q_{2K}	"	"	"	"	17	"	"	"	"
S28	Q_{2L}	"	"	"	"	37	"	"	"	"
S29	Q_{2M}	"	"	"	"	20	"	"	"	"
S30	Q_{2N}	"	"	"	"	40	"	"	"	"
S31	Q_{2O}	"	"	"	"	19	"	"	"	"
S32	Q_{2P}	"	"	"	"	39	"	"	"	"

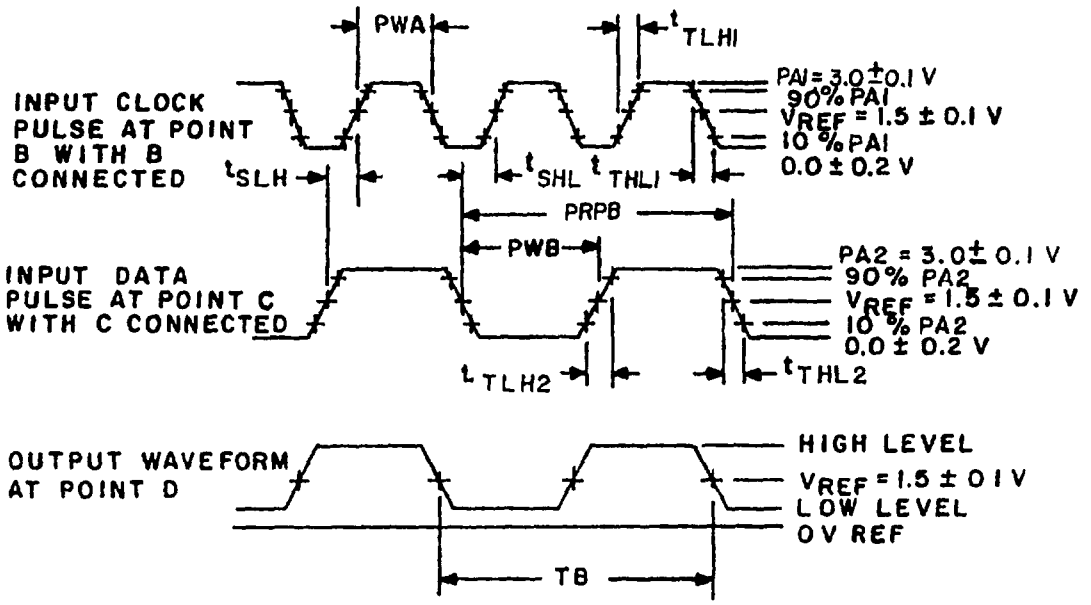
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. Load each output with load as specified on figure 16.

FIGURE 25. Maximum shift frequency test circuit for table XIX.

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NOTES:
1. Input pulse parameters for item J01-1:
   Put item J01-1 in the double pulse mode.
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns.
   PWA = 20 ±2 ns.
2. Input pulse parameters for item J01-2:
   $t_{TLH2} = t_{THL2} = 4 \pm 1$ ns.
   PWB = 40 ±4 ns.
   PRPB = 80 ±8 ns.
3. All voltages are referenced to ground.
4. $t_{SLH} = t_{SHL} = 30 \pm 3$ ns.
5. Verify that all outputs have a high level $\geq$ 2.4 V and a low level $\leq$ 0.4 V before making any measurements.
6. The physical length of electrical conductors from resistors R1 and R2 to input pins, from items Y01-1 and Y01-2 to input pins, and from load circuits to output pins shall be less than 1 inch (25.4 mm).

* FIGURE 26. Maximum shift frequency test circuit waveforms for table XIX.

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TABLE XX. Output crosstalk (low-level) test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T1	Q_{1A}	Output crosstalk, low level	36	5	16	11	24	0	400	mV	27,28
T2	Q_{1B}	"	"	"	"	"	22	"	"	"	"
T3	Q_{1C}	"	"	"	"	"	2	"	"	"	"
T4	Q_{1D}	"	"	"	"	"	21	"	"	"	"
T5	Q_{1E}	"	"	"	"	"	25	"	"	"	"
T6	Q_{1F}	"	"	"	"	"	23	"	"	"	"
T7	Q_{1G}	"	"	"	"	"	3	"	"	"	"
T8	Q_{1H}	"	"	"	"	"	28	"	"	"	"
T9	Q_{1I}	"	"	"	"	"	7	"	"	"	"
T10	Q_{1J}	"	"	"	"	"	27	"	"	"	"
T11	Q_{1K}	"	"	"	"	"	6	"	"	"	"
T12	Q_{1L}	"	"	"	"	"	26	"	"	"	"
T13	Q_{1M}	"	"	"	"	"	29	"	"	"	"
T14	Q_{1N}	"	"	"	"	"	9	"	"	"	"
T15	Q_{1O}	"	"	"	"	"	30	"	"	"	"
T16	Q_{1P}	"	"	"	"	"	31	"	"	"	"

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TABLE XX. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T17	Q_{2A} Output crosstalk, low level	5	36	8	4	32	0	400	mV	27,28
T18	Q_{2B} "	"	"	"	"	12	"	"	"	"
T19	Q_{2C} "	"	"	"	"	33	"	"	"	"
T20	Q_{2D} "	"	"	"	"	13	"	"	"	"
T21	Q_{2E} "	"	"	"	"	15	"	"	"	"
T22	Q_{2F} "	"	"	"	"	35	"	"	"	"
T23	Q_{2G} "	"	"	"	"	14	"	"	"	"
T24	Q_{2H} "	"	"	"	"	34	"	"	"	"
T25	Q_{2I} "	"	"	"	"	18	"	"	"	"
T26	Q_{2J} "	"	"	"	"	38	"	"	"	"
T27	Q_{2K} "	"	"	"	"	17	"	"	"	"
T28	Q_{2L} "	"	"	"	"	37	"	"	"	"
T29	Q_{2M} "	"	"	"	"	20	"	"	"	"
T30	Q_{2N} "	"	"	"	"	40	"	"	"	"
T31	Q_{2O} "	"	"	"	"	19	"	"	"	"
T32	Q_{2P} "	"	"	"	"	39	"	"	"	"

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TABLE XXI. Output crosstalk (high-level) test.

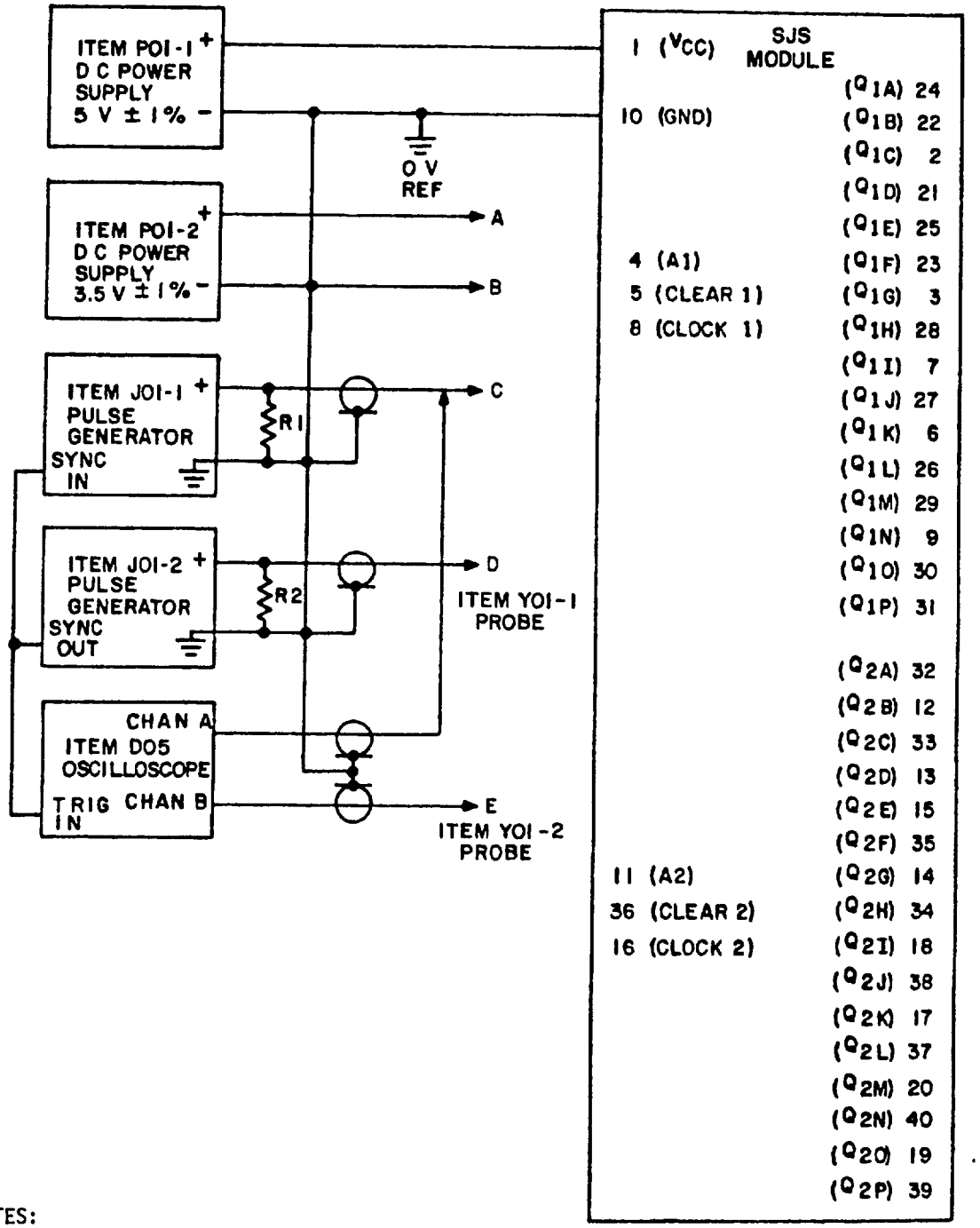
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
U1	Q_{1A}	Output crosstalk, high level	5,36,4	Open	8,16	11	24	0	400	mV	27,28
U2	Q_{1B}	"	"	"	"	"	22	"	"	"	"
U3	Q_{1C}	"	"	"	"	"	2	"	"	"	"
U4	Q_{1D}	"	"	"	"	"	21	"	"	"	"
U5	Q_{1E}	"	"	"	"	"	25	"	"	"	"
U6	Q_{1F}	"	"	"	"	"	23	"	"	"	"
U7	Q_{1G}	"	"	"	"	"	3	"	"	"	"
U8	Q_{1H}	"	"	"	"	"	28	"	"	"	"
U9	Q_{1I}	"	"	"	"	"	7	"	"	"	"
U10	Q_{1J}	"	"	"	"	"	27	"	"	"	"
U11	Q_{1K}	"	"	"	"	"	6	"	"	"	"
U12	Q_{1L}	"	"	"	"	"	26	"	"	"	"
U13	Q_{1M}	"	"	"	"	"	29	"	"	"	"
U14	Q_{1N}	"	"	"	"	"	9	"	"	"	"
U15	Q_{1O}	"	"	"	"	"	30	"	"	"	"
U16	Q_{1P}	"	"	"	"	"	31	"	"	"	"

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TABLE XXI. Output crosstalk (high-level) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
U17	Q_{2A}	Output crosstalk, high level	11,36,5	Open	8,16	4	32	0	400	mV	27,28
U18	Q_{2B}	"	"	"	"	"	12	"	"	"	"
U19	Q_{2C}	"	"	"	"	"	33	"	"	"	"
U20	Q_{2D}	"	"	"	"	"	13	"	"	"	"
U21	Q_{2E}	"	"	"	"	"	15	"	"	"	"
U22	Q_{2F}	"	"	"	"	"	35	"	"	"	"
U23	Q_{2G}	"	"	"	"	"	14	"	"	"	"
U24	Q_{2H}	"	"	"	"	"	34	"	"	"	"
U25	Q_{2I}	"	"	"	"	"	18	"	"	"	"
U26	Q_{2J}	"	"	"	"	"	38	"	"	"	"
U27	Q_{2K}	"	"	"	"	"	17	"	"	"	"
U28	Q_{2L}	"	"	"	"	"	37	"	"	"	"
U29	Q_{2M}	"	"	"	"	"	20	"	"	"	"
U30	Q_{2N}	"	"	"	"	"	40	"	"	"	"
U31	Q_{2O}	"	"	"	"	"	19	"	"	"	"
U32	Q_{2P}	"	"	"	"	"	39	"	"	"	"

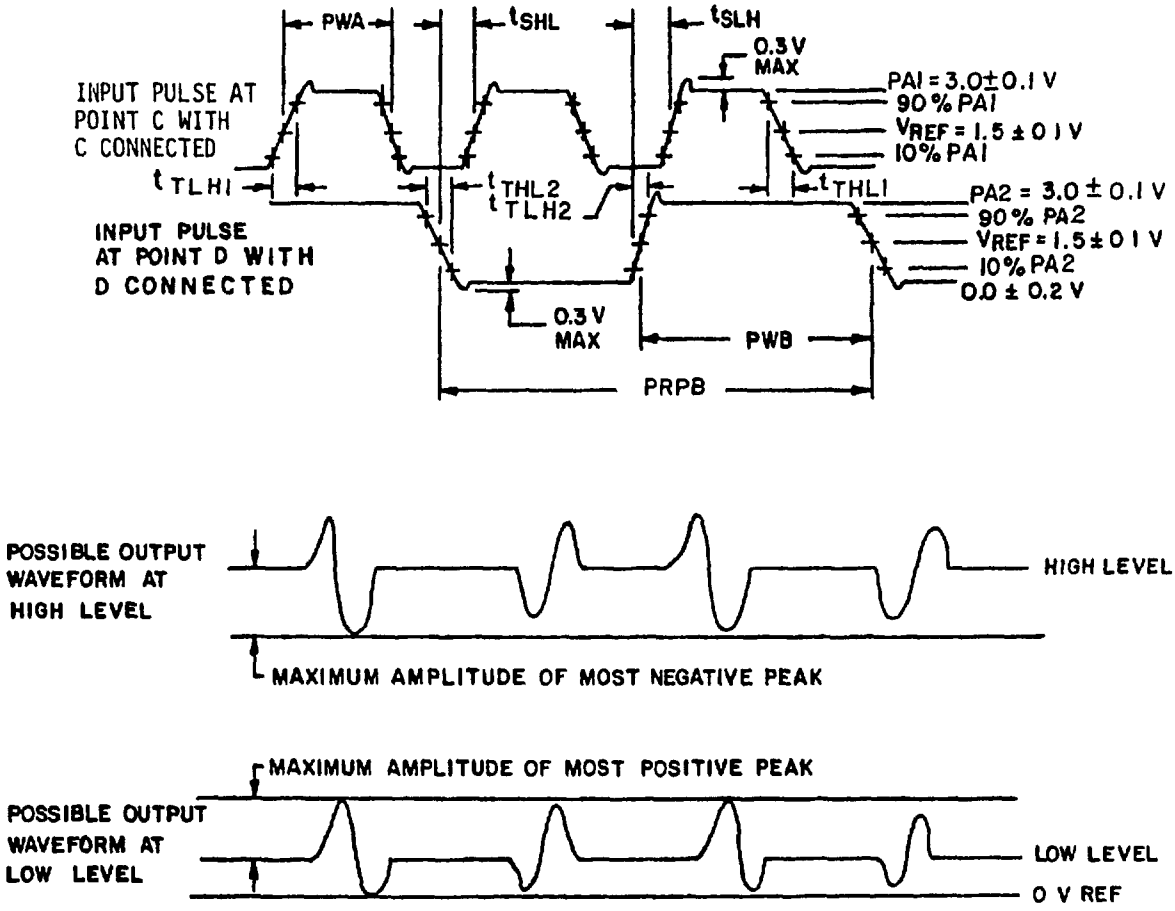
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. Load each output with load as specified on figure 16.
3. Allow a minimum of 17 clock pulses before making measurements for test condition T1-T32.

FIGURE 27. Output crosstalk (low-level and high-level) test circuit for tables XX and XXI.

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NOTES:
1. Input pulse parameters for item J01-1 (verified at each input pin):
   PWA = 50 ±5 ns.
   $t_{TLH1} = t_{THL1} = 5 \pm 1$ ns.
   Put item J01-1 in the double pulse mode.
2. Input pulse parameters for item J01-2 (verified at each input pin):
   PWB = 100 ±10 ns.
   PRPB = 200 ±20 ns.
   $t_{TLH2} = t_{THL2} = 10 \pm 2$ ns.
3. $t_{SHL} = t_{SLH} = 20 \pm 2$ ns.
4. Verify the output voltages low level and high levels are $\leq$ 0.4 V and $\geq$ 2.4 V respectively.
5. All voltages are referenced to ground.

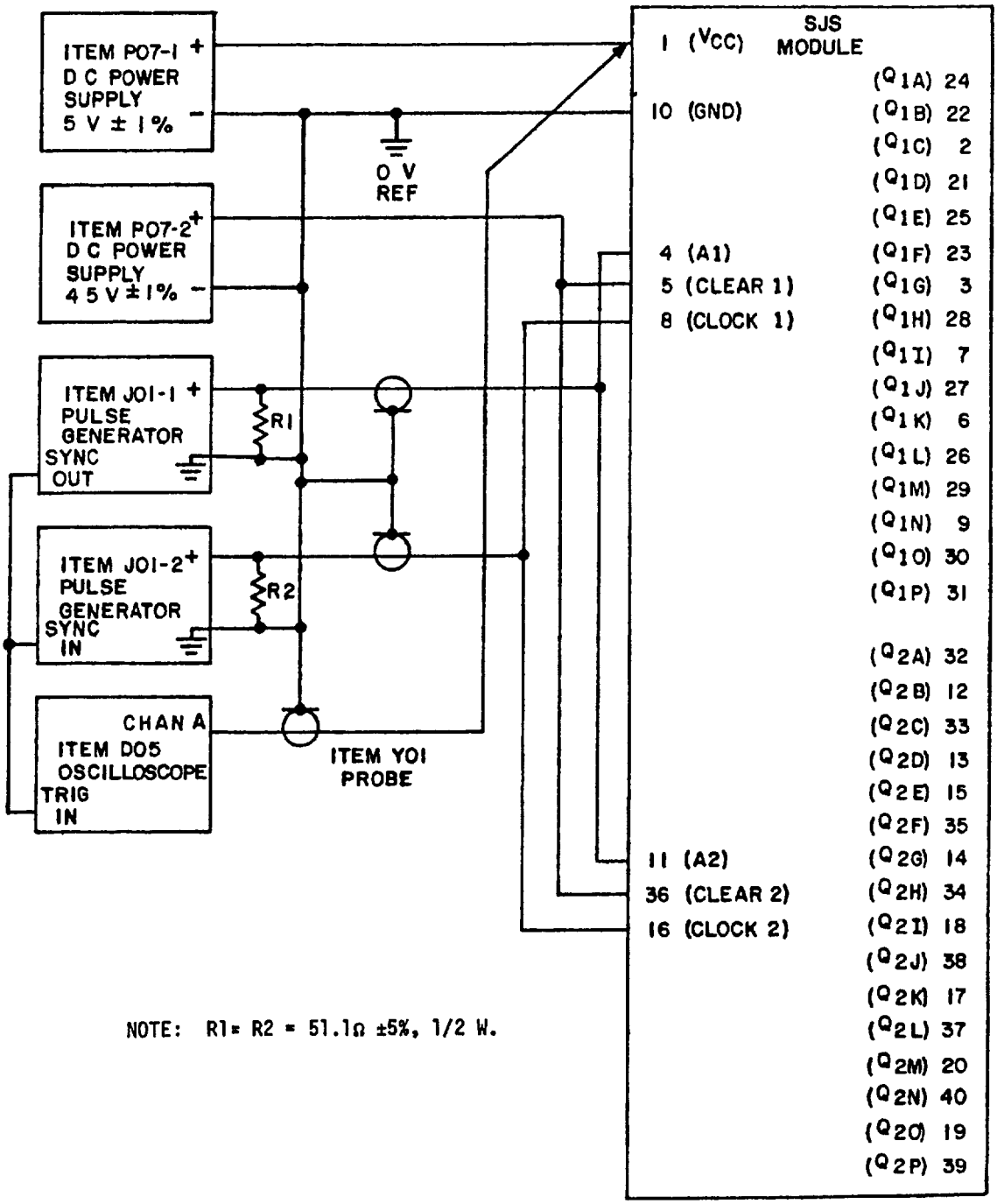
* FIGURE 28. Output crosstalk (low-level and high-level) test circuit waveform for tables XX and XXI.

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TABLE XXII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
V	Power supply transient amplitude	0	400	mV	29, 30

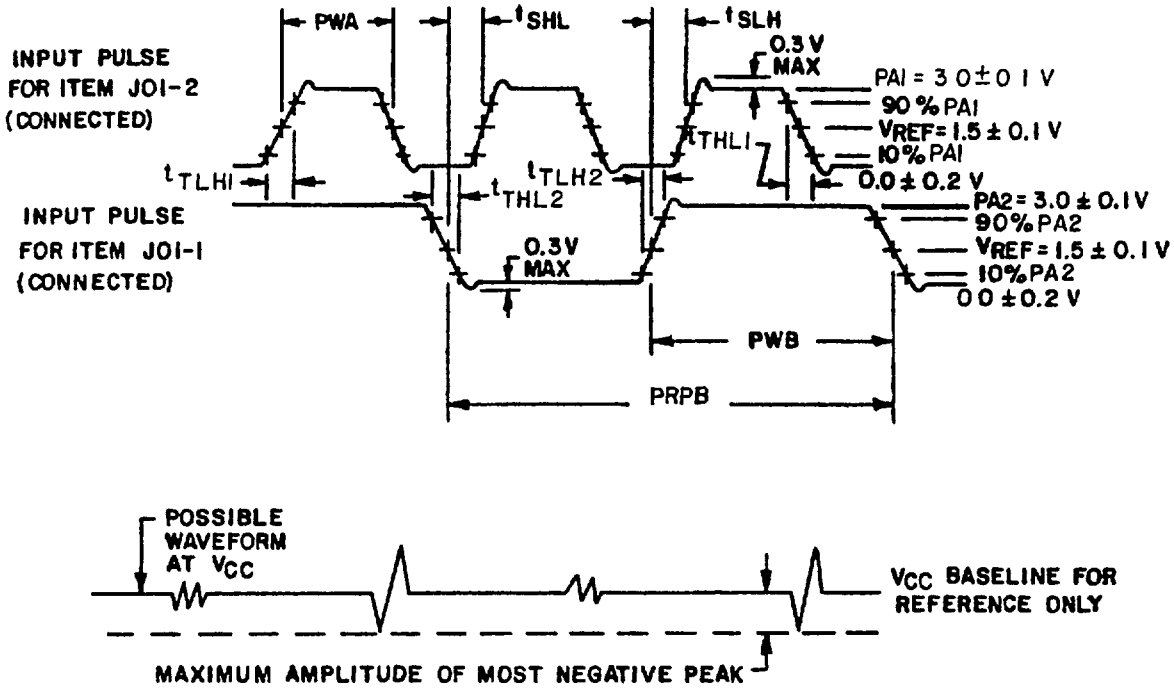
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NOTE: R1= R2 = 51.1Ω ±5%, 1/2 W.

FIGURE 29. Power supply transient amplitude test circuit for table XXII.

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NOTES:
1. Input pulse parameters for item J01-2 (verified at each input pin):
   PWA = 250 ±25 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   Put item J01-1 in the double pulse mode.
2. Input pulse parameters for item J01-1 (verified at each input pin):
   PWB = 500 ±50 ns.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
   PRPB = 1,000 ±100 ns.
3. $t_{SHL} = t_{SLH}$ = 200 ±20 ns.
4. Verify the output voltages low level and high levels are $\leq$ 0.4 V and $\geq$ 2.4 V respectively.
5. All voltages are referenced to ground.

FIGURE 30. Power supply transient amplitude test circuit waveforms for table XXII.

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TABLE XXIII. Power supply filter capacitance test.

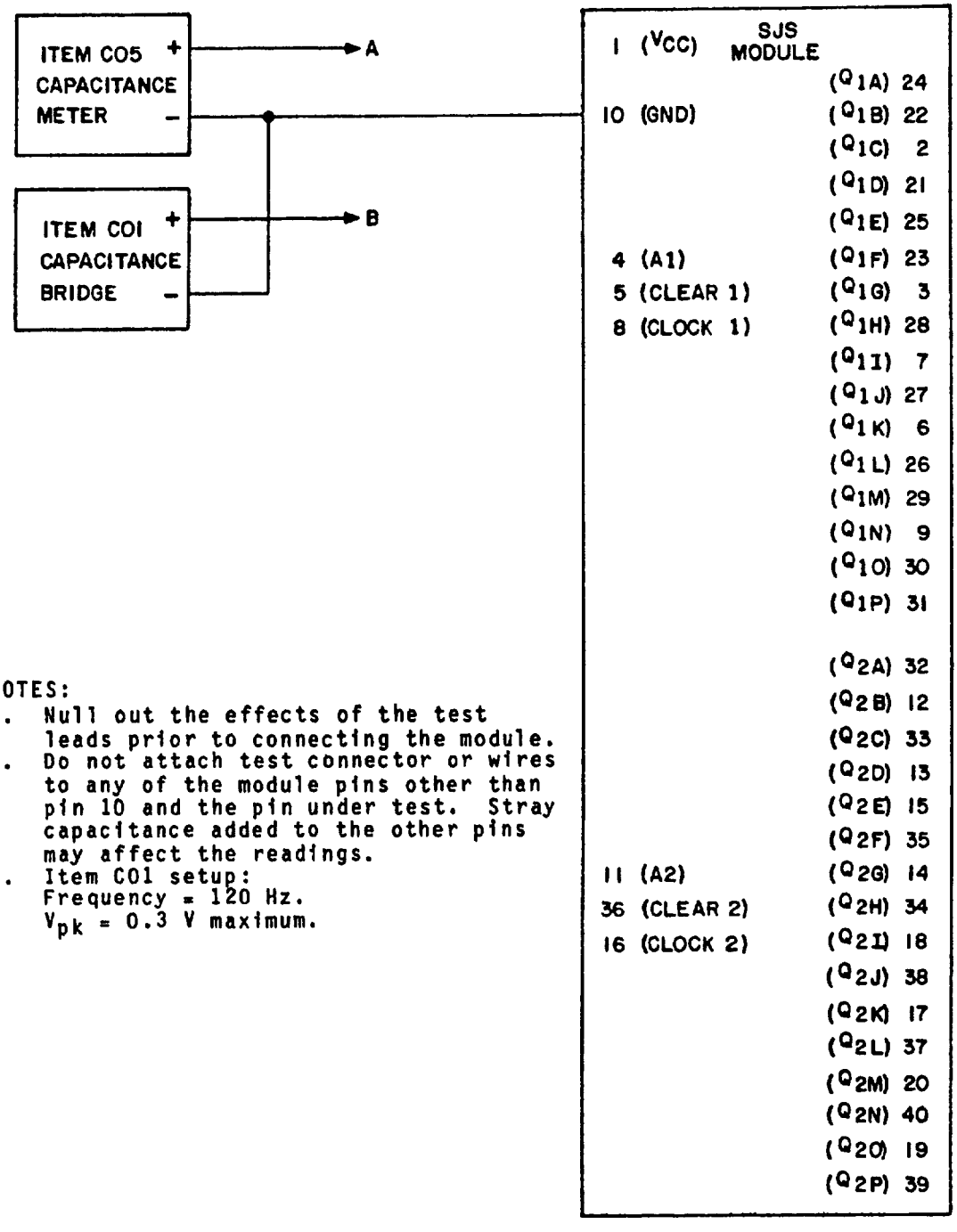
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W	Power supply filter capacitance	Open	1	3.30	6.12	µF	31

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TABLE XXIV. Input capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
X1	A1	Input capacitance	4	Open	2	30	pF	31
X2	CLEAR 1	"	5	"	"	"	"	"
X3	CLOCK 1	"	8	"	"	"	"	"
X4	A2	"	11	"	"	"	"	"
X5	CLEAR 2	"	36	"	"	"	"	"
X6	CLOCK 2	"	16	"	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to the other pins may affect the readings.
3. Item COl setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

* FIGURE 31. Power supply filter capacitance and input capacitance test circuit for tables XXIII and XXIV.

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